OliverWyman

ST. JOHNSBURY HSA
NORTHEASTERN VERMONT REGIONAL HOSPITAL

Act 167 (2022) Community Engagement to Support Hospital Transformation

Aug 2024

A business of Marsh McLennan

TABLE OF CONTENT

1. ACT 167 (2022)

SECTION 2 CONTEXT

PROJECT CONTEXT

Act 167 objectives

Section 2 of Act 167 (2022) requires that the GMCB, in collaboration with the Director of Health Care Reform in the Agency of Human Services, “develop and conduct a data-informed, patient-focused, community-inclusive engagement process for Vermont’s hospitals to:

- **Reduce inefficiencies**
- **Lower costs**
- **Improve population health outcomes**
- **Reduce health inequities**
- **Increase access to essential services**

All while maintaining sufficient capacity for emergency management

Oliver Wyman’s work

Broad community and provider engagement in and across all Hospital Service Areas (HSAs) in Vermont.

– a data-informed, patient-focused, community-inclusive engagement relative to the second stream of work for Act 167

Goals: engage diverse stakeholders and ascertain their interactions with the health system and perceived needs to overcome any barriers to equitable access and outcomes

Project plan: *conducted interviews and working sessions in 2 phases, and used data analyzed in Phase 2a to help codify quantitative and qualitative inputs to inform a recommendation.*

– *Phase 1 - August through mid-November 2023*
– *Phase 2a- November 2023 through March 2024*
– *Phase 2b- March through September 2024*
– *Timing of the phases are subject to change by mutual agreement*

2. WORK TO DATE

SCOPE AND APPROACH: IMPROVING THE VERMONT HEALTHCARE DELIVERY SYSTEM REQUIRES INPUT FROM ACROSS THE COMMUNITY IT SERVES

Act 167 (of 2022) requires GMCB, in collaboration with the Agency of Human Services, to develop and conduct a data-informed, patient-focused, community-inclusive engagement process for Vermont's hospitals to **reduce inefficiencies, lower costs, improve population health outcomes, reduce health inequities, and increase access to essential services**

Green Mountain Care Board:
Community and Provider Engagement

Agency of Human Services:
System Transformation

Aug-Sept 2023

Step 1. Finalize engagement plan and interview scheduling

Identify and schedule interviews with key stakeholders:

- Hospital leaders
- Provider organizations
- Community leaders and members
- Diverse populations
- State of Vermont partners and Legislators
- Health related organizations

Sept-Nov 2023

Step 2. Develop current state understanding in 1st round interviews

Obtain community perspective on:

- Hospital Service Area (HSA) characteristics
- Hospital & healthcare delivery system performance
- Community needs
- Desired health system future state

March – July 2024

Step 3. Develop solution options, vet with community

- Develop solutions to address current needs and reach the desired future state while considering Act 167 goals
  - *In conjunction with analytics contractor*
- Obtain stakeholder perspective on the impact of recommendations on hospital performance and healthcare delivery

Completed Phases

Aug – Sep 2024

Step 4. Develop and deliver final report

- Document and socialize confirmed current state understanding, designed future state, and recommended steps to achieve future state (including pros and cons)
- Publish report

Sept 2024 - ?

Step 5. Transform healthcare delivery system

- Devise action plan and budget allocations
- Implement chosen healthcare delivery system transformations

ST. JOHNSBURY SUMMARY
HOSPITAL: NORTHEASTERN VERMONT REGIONAL HOSPITAL

NVRH noted they provide Community Health Worker support to meet transportation needs. Providers/our system can refer individuals to our Community Connections program for support. NVRH also maintains a donor-supported unmet needs fund to support the cost.

Bright spots

Internal hospital/provider operations	• *Expected $30m expansion of ED and lab facilities, waiting on certificate of need* • Providers are overburdened with administrative tasks that take up a lot of patient-facing time
Coordination between organizations	• *Some coordination exists to ensure access to specialists and facilities for patients (i.e., sleep lab share with North Country Hospital, tele-neurology with Dartmouth, UVMMC for cardiac services)* • Interoperability is an issue broadly across the state with current EMR systems • Lacks a system to streamline referrals and shared care planning in the team-based care like UniteUs • Culture, weather, and geography are major deterrents for coordinating shared services with hospitals in Newport
Transport and infrastructure	• Patients will forego treatment given long distance to specialists or other facilities (i.e. dialysis, dermatology) • *They have a "rides to wellness program" with a local transportation provider (RCT*), but there are challenges in finding drivers • Shortage of staff to enable proper transfers to other facilities causes delays and adds to expenses • EMS and home health are understaffed and use old equipment; hospital struggles to meet community needs
Work force	• Attracting and retaining nurses and providers are challenging on several fronts – Housing is an issue to getting providers into the facilities, and high cost of living makes it difficult to retain providers – Hospital pays Boston wages to attract staff, but it drives up cost – High burnout rate given providers are over-burned in certain practices (ED, primary care) • *Implemented / invested ($500K) in a career advancement program – aimed at growing local nursing and clinical professionals* • Violence in the ED is common; there is need for law enforcement/security
Financials *excluding workforce*	• Telehealth initiatives with Dartmouth are costly (annual fee, set up fees, costs them $600 for consult but are reimbursed $150/consult)
Patient centered care *Culturally competent / equitable care*	• There is subset of patients that stay in the hospital because they do not have alternative housing or care options (i.e., mental health) leading to high costs and lower bed availability • Community health work force is critical for ensuring community health services, but it doesn't generate revenue and there isn't enough margin to make further investments • Poverty issue the main issue impacting equity locally (e.g., internet access, electricity, inability to contact patients)
Services offered	• Noted services needed: psychiatry, pharmacy services close to the hospital as most closed, dental/vision • *Leader for VT healthcare reform (i.e., pilot site for VT Blueprint for Health, leverages PCMH, and participates in OneCare VT)* • Facilities and providers are insufficient to treat the increasing aging population and their complex medical needs • Persistent and long wait times for patients to access any care

Source: Hospital and community meetings (August- November 2023)

3. HSA VIEW

ST. JOHNSBURY

IN THE HSA VIEW, WE WILL DISCUSS

Population change

- Overall population trends
- Age break-down

Trends in healthcare needs and service demand

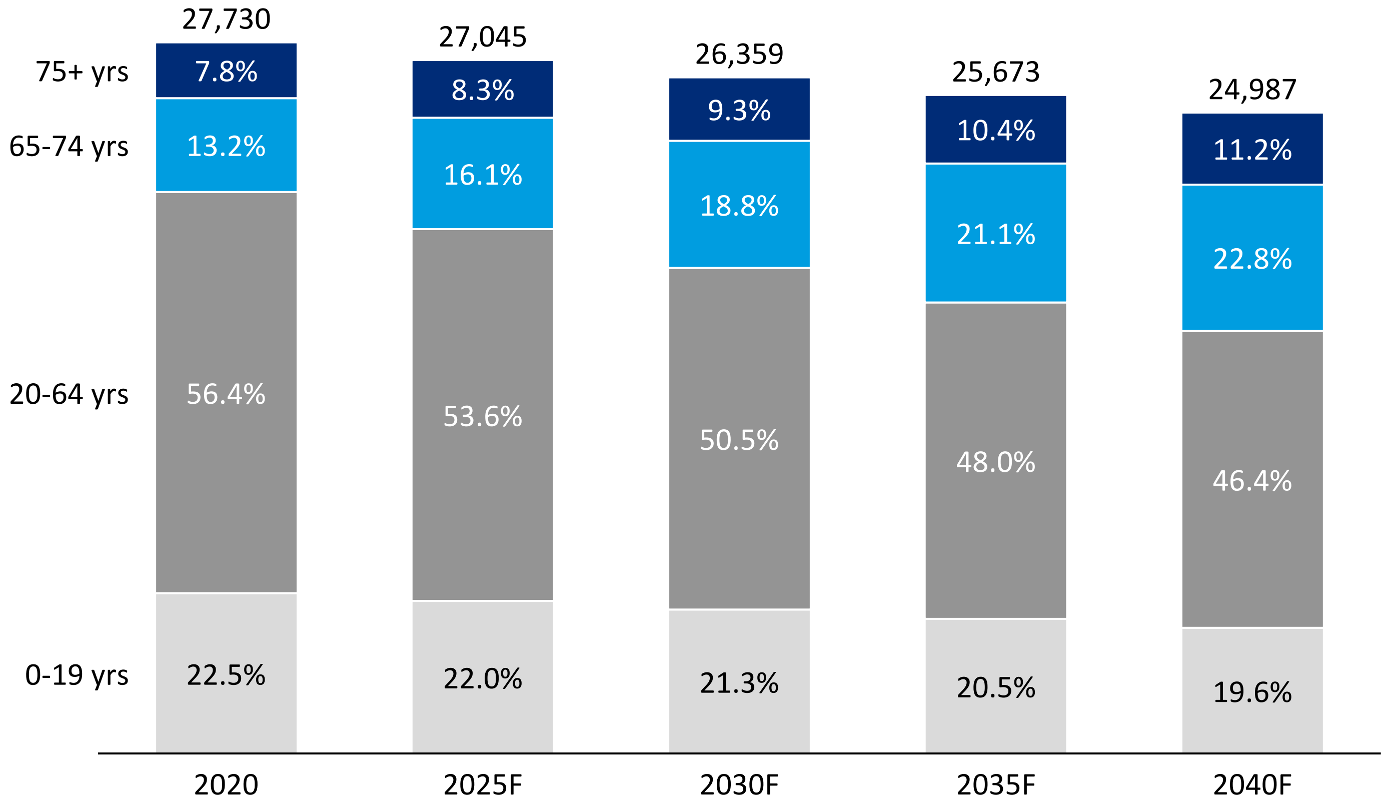
- Cancer
- Cardiovascular
- Obstetrics
- PCP/IP/ED/OP

Current and projected resource capacity

- PCP, APRN, mental health counselors
- IP bed, Nursing home beds, mental health beds

ST. JOHNSBURY HSA IS FORECASTED TO HAVE AN INCREASINGLY AGING POPULATION WITH SHRINKING POPULATION POOL

Projected St. Johnsbury population break-down[1]
2020-2040F



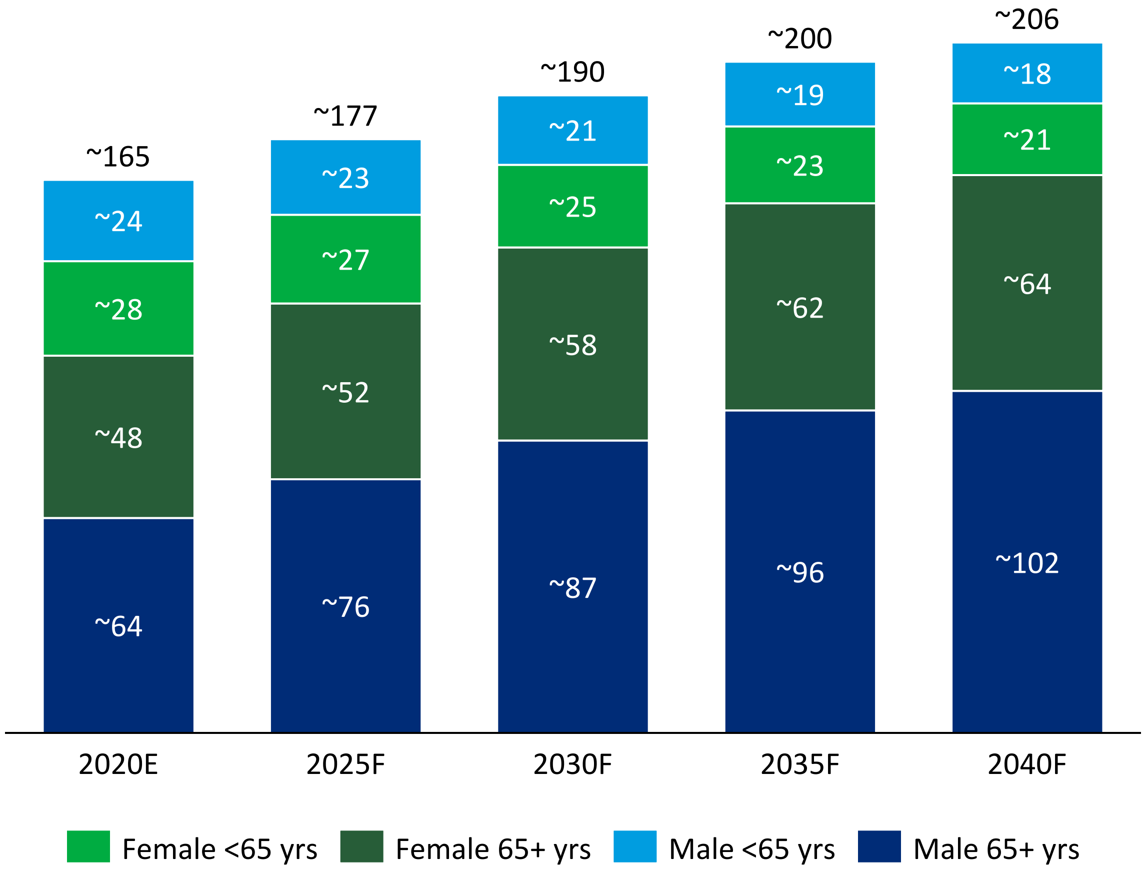
Population trends and implications

- Proportion of 65+ year old residents in St. Johnsbury HSA is forecasted to increase from 21.0% in 2020 to 34.0% in 2040
- Between 2020 and 2040, Medicare eligible population (65+ yrs) is forecasted to increase by 45.8%, whilst working age population (20-64 yrs) will decrease by 26.0%, shifting payer mix towards Medicare significantly and **rendering cross-subsidization increasingly unsustainable as a financial strategy**
- Need for support services such as social services, nursing home, transportation will be greater as patient population ages

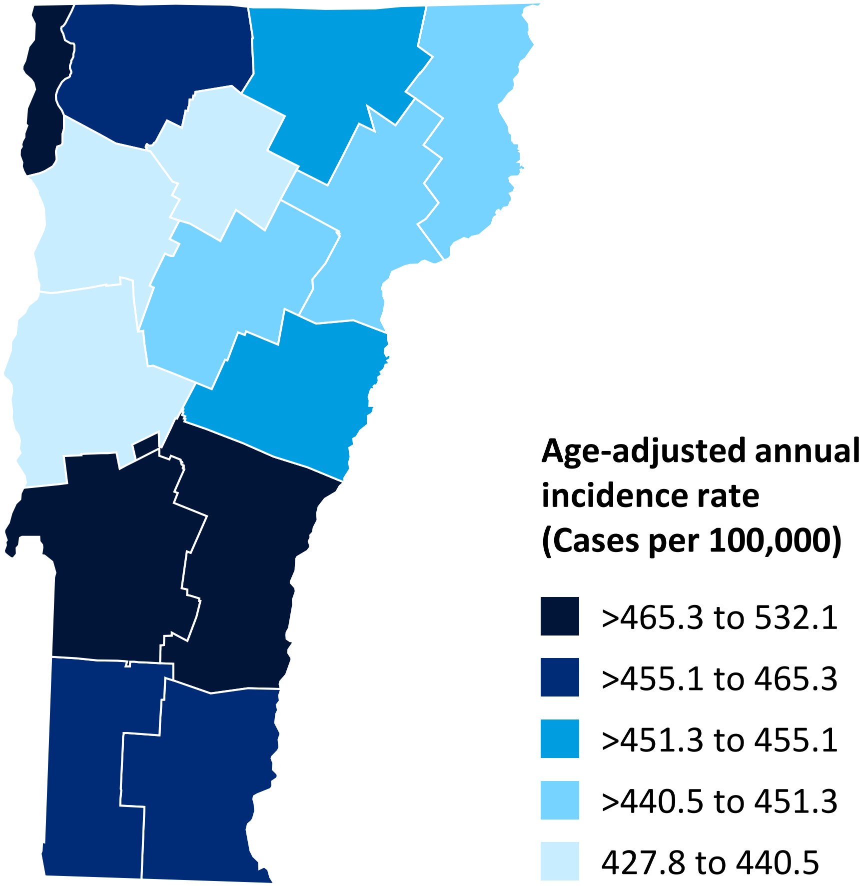
Source: 1. MPR VT Population by HSA, Oliver Wyman analysis

CANCER CASES ARE EXPECTED TO INCREASE MODESTLY AS POPULATION AGES

Forecasted new cancer cases in St. Johnsbury by sex and age
2020F-2040F



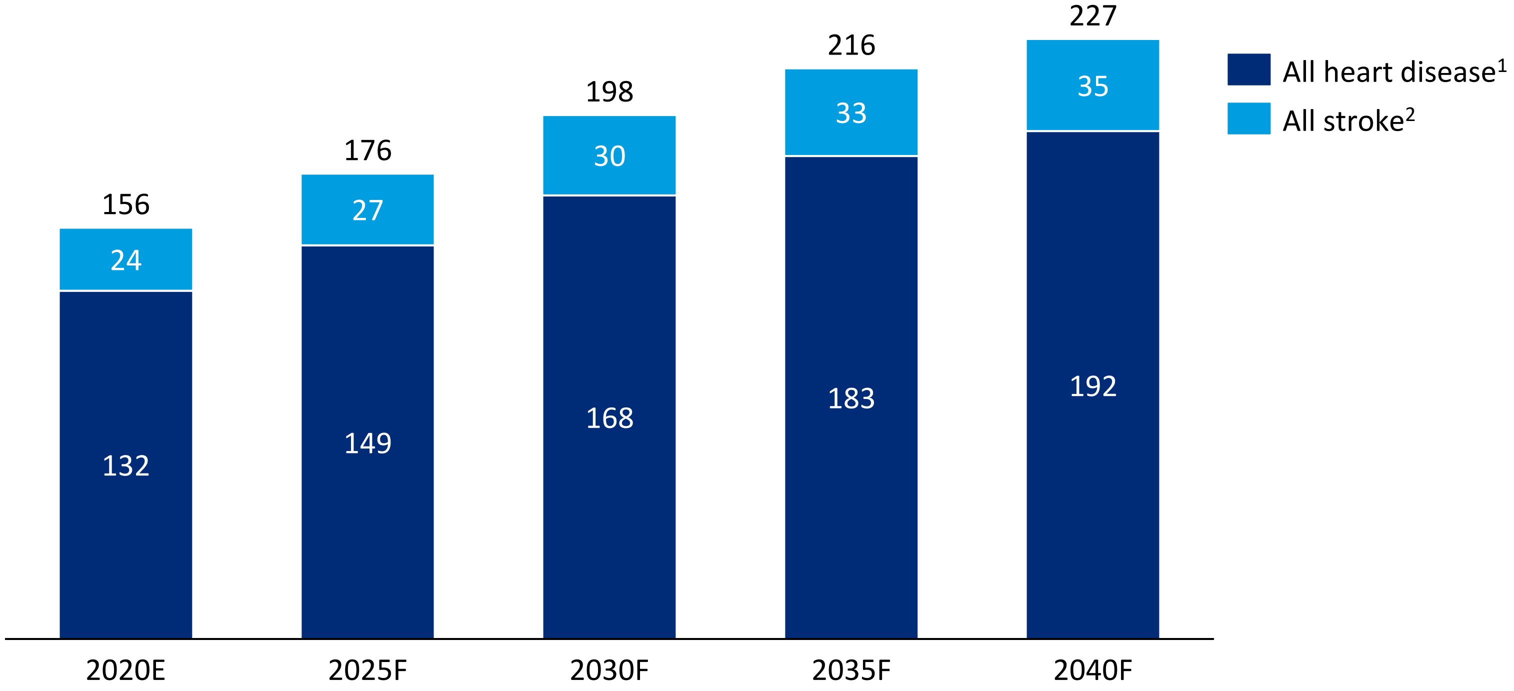
Cancer incidence rate in Vermont, by county
All cancer sites, 2016-2020



Source: MPR VT Population by HSA, NIH State Cancer Profiles Incidence Rate Report for Vermont by County in 2016-2020 (link), Oliver Wyman analysis

NUMBER OF HEART DISEASE OR STROKE RELATED HOSPITALIZATIONS AMONG SENIORS IS ALSO PREDICTED TO INCREASE AS POPULATION AGES

Forecasted number of hospitalizations of 65+yrs individuals due to cardiovascular disease or stroke
St. Johnsbury, 2020E-2040F



Remarks

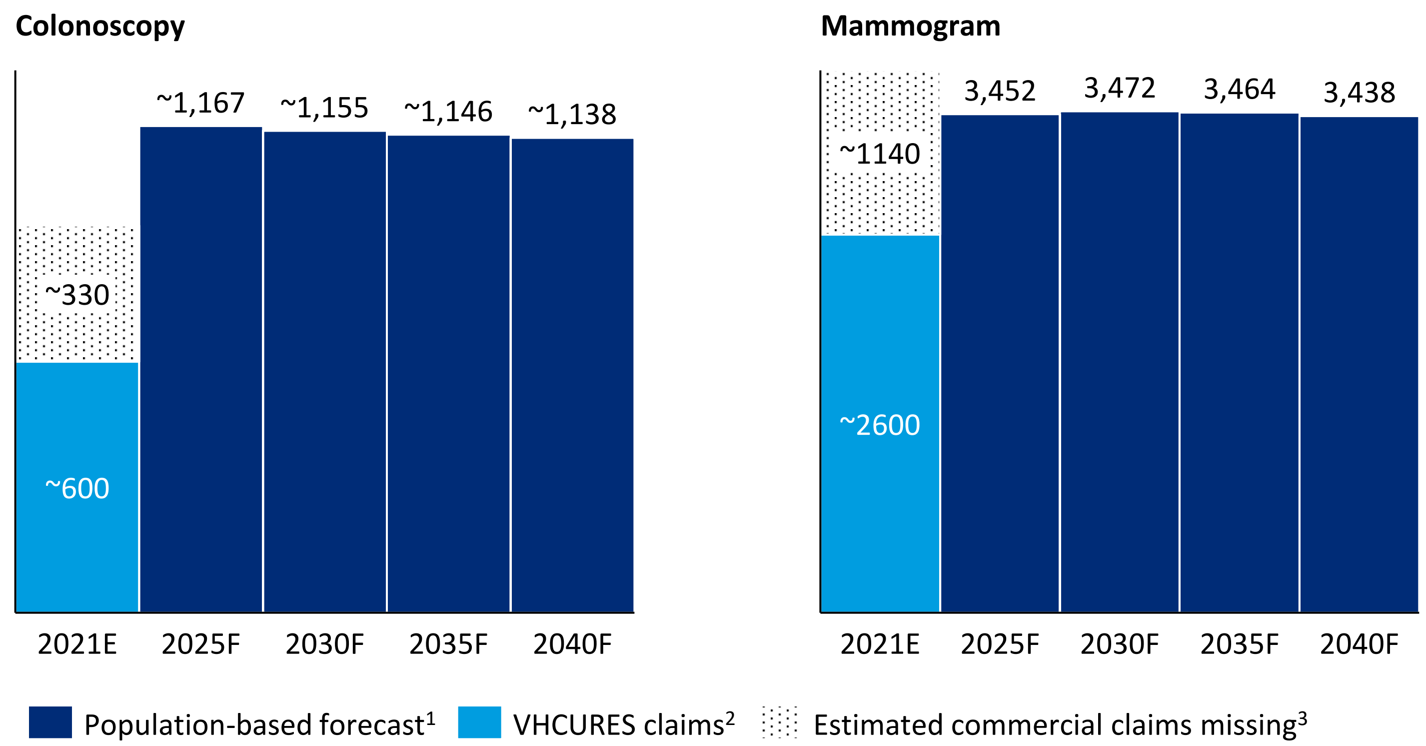
- Proportion of 65+ year old residents in St. Johnsbury HSA is forecasted to increase from 21.0% in 2020 to 34.0% in 2040
- Cardiovascular-related cases increase 50% with most in Heart Disease

1. Calculated by multiplying all heart disease hospitalization rate per 1,000 Medicare beneficiaries (all races/ethnicities, all genders, ages 65+, 2019-2021) in Caledonia County to forecasted population of 65+ yrs in St. Johnsbury
2. Calculated by multiplying all stroke hospitalization rate per 1,000 Medicare beneficiaries (all races/ethnicities, all genders, ages 65+, 2019-2021) in Caledonia County to forecasted population of 65+ yrs in St. Johnsbury

Source: CDC Interactive Atlas of Heart Disease and Stroke as accessed on 16 April, MPR VT Population by HSA, Oliver Wyman analysis

UTILIZATION OF COLONOSCOPY AND MAMMOGRAM IS FORECASTED TO REMAIN STABLE

Colonoscopy and mammogram utilization in St. Johnsbury
2021E, 2025F-2040F



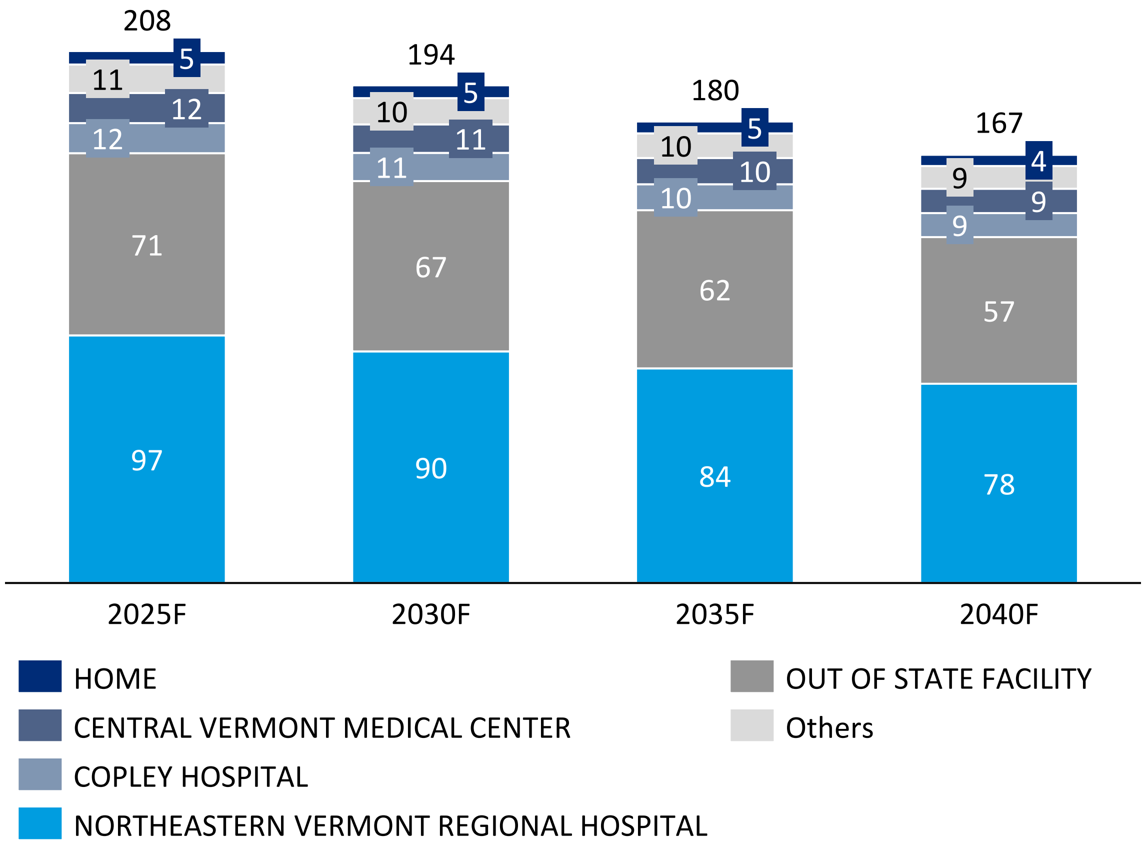
Remarks

- Demand for colonoscopy and mammograms is forecasted to remain at a stable level as increasing healthcare needs by elderly population is balanced by shrinking population pool

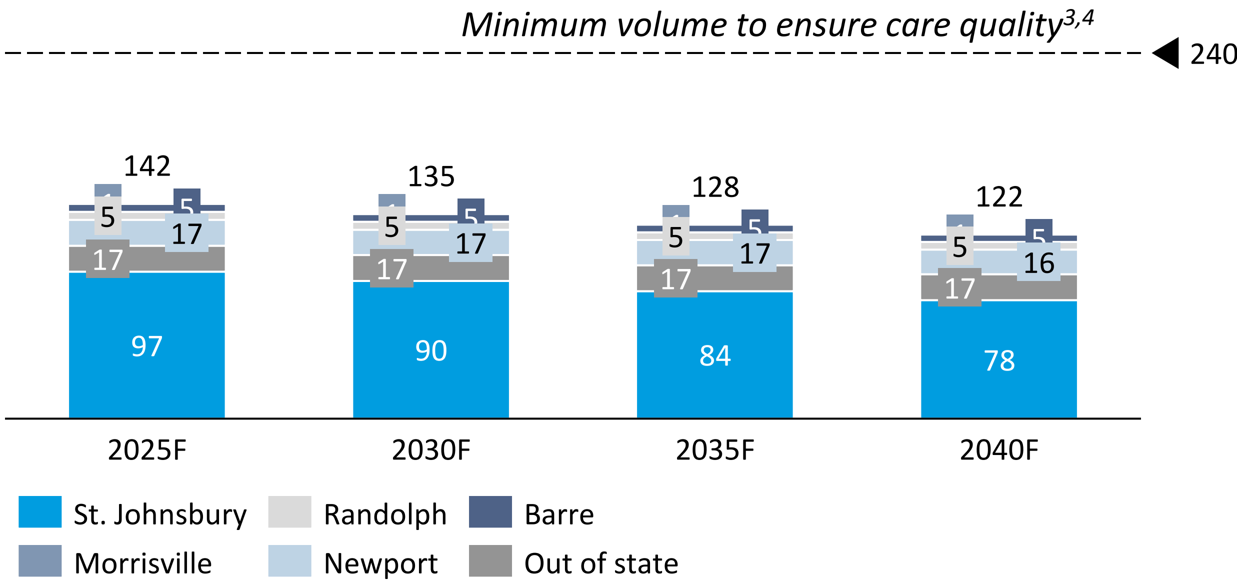
1. Assuming USPSTF recommendations: colonoscopy every 10 years for individuals aged 45 to 74, breast cancer screening every 2 years for women aged 40 to 74, 2. VHCURES claims has incomplete coverage of claims, missing those from private health insurance companies and private sector self-insured employers that choose not to submit to VHCURES, out-of-pocket care, individuals covered by military and federal employee health plans and payers with very few Vermont residents (less than 200) enrolled. 3. Assuming half of commercial claims are missing from VHCURES data
Source: 2021 VHCURES claims, MPR VT Population by HSA, Oliver Wyman analysis

EXPECTED DELIVERIES IN NVRH ARE LIKELY TO DECREASE, UNLESS NVRH CAN RETAIN MORE BIRTHS ORIGINATING FROM ITS HOME HSA

St. Johnsbury births by place of birth[1,2]
2025F-2040F



NVRH births by patient residence[1,2]
2025F-2040F

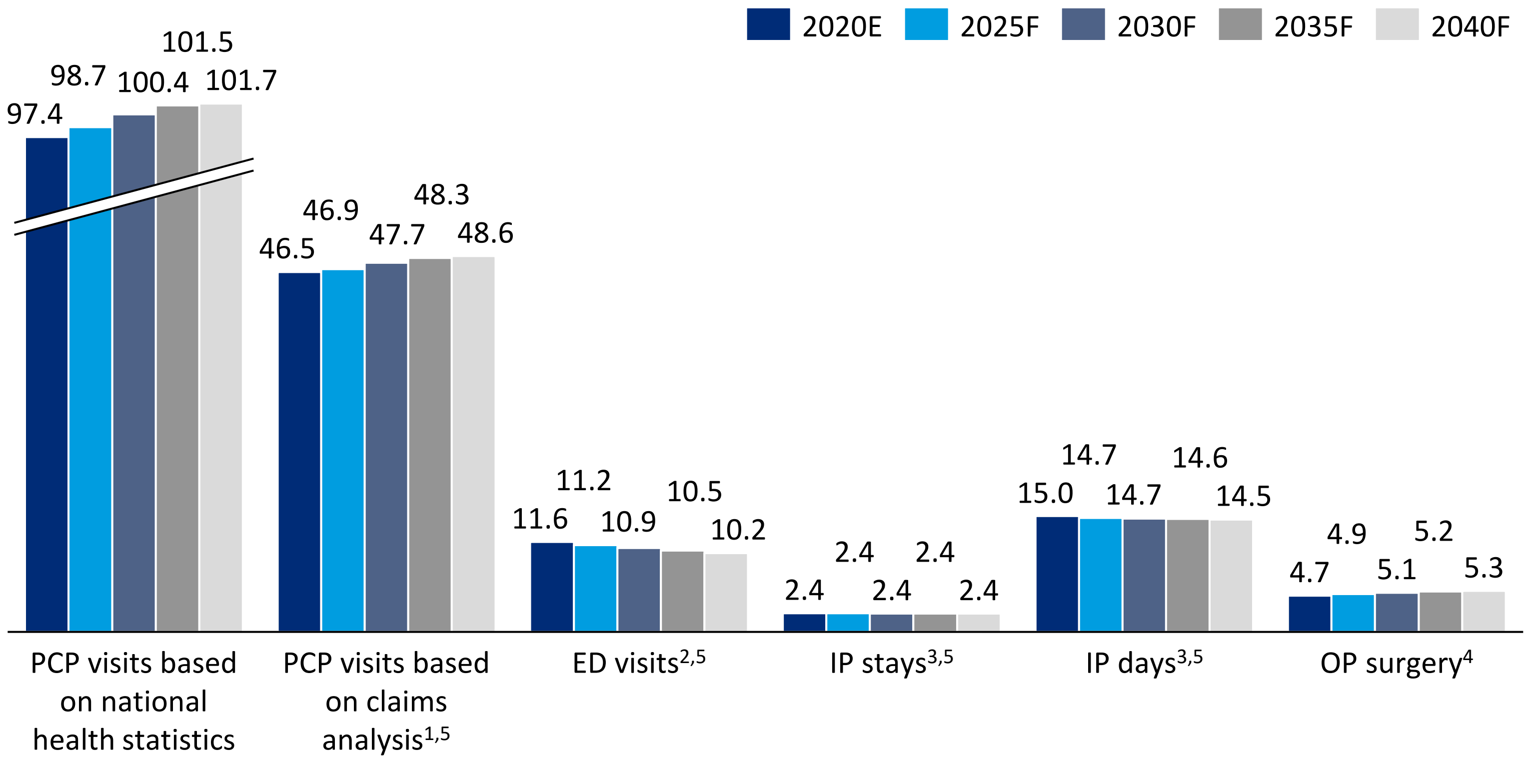


Source: 1. 2021 Vermont Vital Statistics Report (link), 2. MPR VT Population forecast, 3. Obstetric Volume and Severe Maternal Morbidity Among Low-risk and Higher Risk Patients Giving Birth at Rural and Urban US Hospitals, Kozhimannil KB, JAMA Health Forum. 2023;4(6):e232110. doi:10.1001/jamahealthforum.2023.2110, 4. Hospital Level Factors Associated with Anesthesia Related Adverse Events in Cesarean Deliveries, New York State 2009-2011. Guglielminotti J, Anesth. Analges. 2016; 122:1947-56, Oliver Wyman Analysis

BASED ON PROJECTED POPULATION GROWTH, ST. JOHNSBURY WILL HAVE HIGHER DEMAND FOR PCP VISITS AND OP SURGERY AND LESS DEMAND FOR ED VISITS

Forecasted healthcare demand in St. Johnsbury based on population forecast

In thousands, 2020E-2040F



Remarks

- In St. Johnsbury, demand for PCP visits and OP surgeries is forecasted to increase as population ages
  - Better connectivity to community-based organizations will be called for (primary care practices, mobile units, nursing homes etc.)

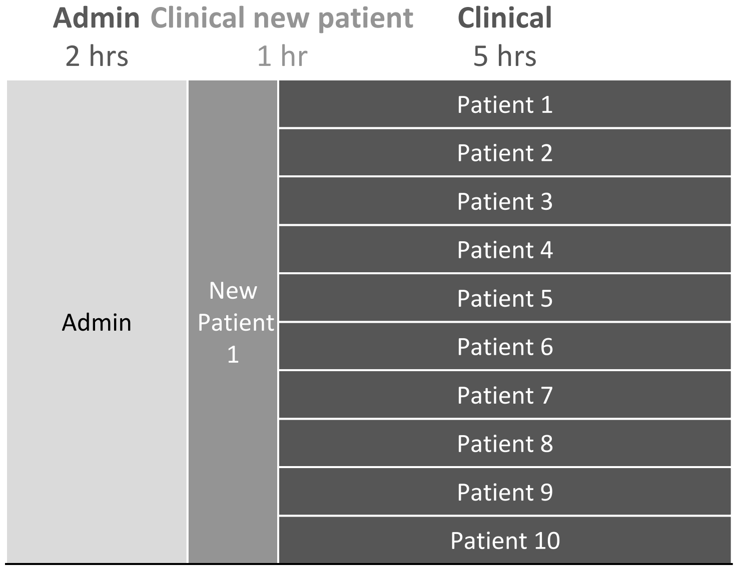
Note: overall level of demand is likely underestimated due to incomplete coverage of VHCURES claims data

1. PCP visits were identified using the same provider taxonomies (that include nurse practitioners and physician assistants) and Current Procedural Terminology (CPT) codes used by GMCB in its report to the Vermont Legislature on primary care spending in accordance with Act 17.1. 2. Emergency Department visits were identified from facility claims for Vermont residents in the VHCURES database at an outpatient acute hospital setting with revenue center codes corresponding to Emergency Department care. Denied claims were omitted. 3. Inpatient stays and days were identified from facility claims for Vermont residents in the VHCURES database at an inpatient or acute care hospital setting. Denied claims and claims with no admission or discharge date were omitted. 4. Results of outpatient surgical demand model, based on VHCURES and VHUDDS data 2016–2019, based on demographic trends only. 5. Assuming age-specific visit or stay rates remain constant as the average of 2021 and 2022 level

Source: Commercial, Medicaid and Medicare FFS professional and outpatient claims from VHCURES in calendar years 2021-2022 (note: Commercial claims from VHCURES do not include claims to the Federal Employee Health Benefit Plan, TRICARE (formerly known as Champus), some self-insured commercial plans, plans with a commercial insurer that covers very few Vermont residents, and Veteran Affairs). MPR VT Population by HSA. GMCB/MPR outpatient surgical demand model, based on VHCURES and VHUDDS data, 2016–2019. National health statistics on PCP visits (link)

PCP PRODUCTIVITY IS A KEY LEVER IN MEETING COMMUNITY PRIMARY CARE DEMAND

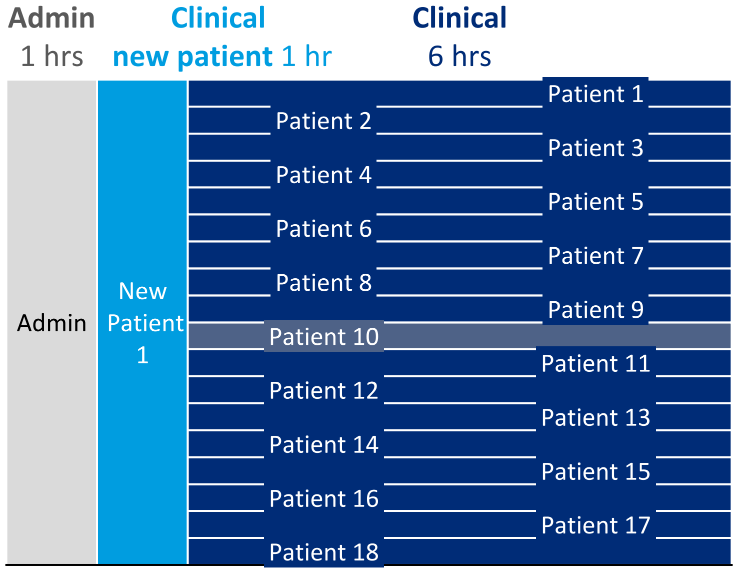
Break-down of PCP time in a working day[1]
30 hrs clinical hours a week, 2 visits per hour



11 patients x 5 working days x 48 working weeks =

2640 PCP visits per year per PCP FTE

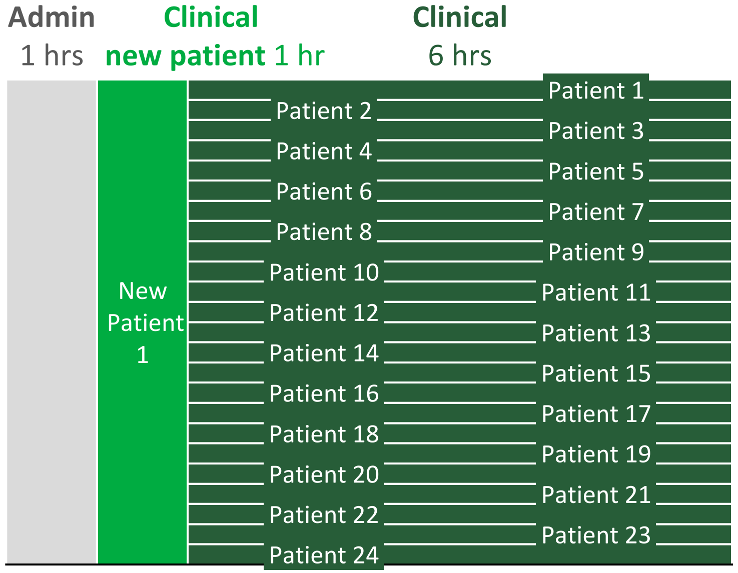
Break-down of PCP time in a working day[1]
35 hrs clinical hours a week, 3 visits per hour



19 patients x 5 working days x 48 working weeks =

4560 PCP visits per year per PCP FTE

Break-down of PCP time in a working day[1]
35 hrs clinical hours a week, 4 visits per hour



25 patients x 5 working days x 48 working weeks =

6000 PCP visits per year per PCP FTE

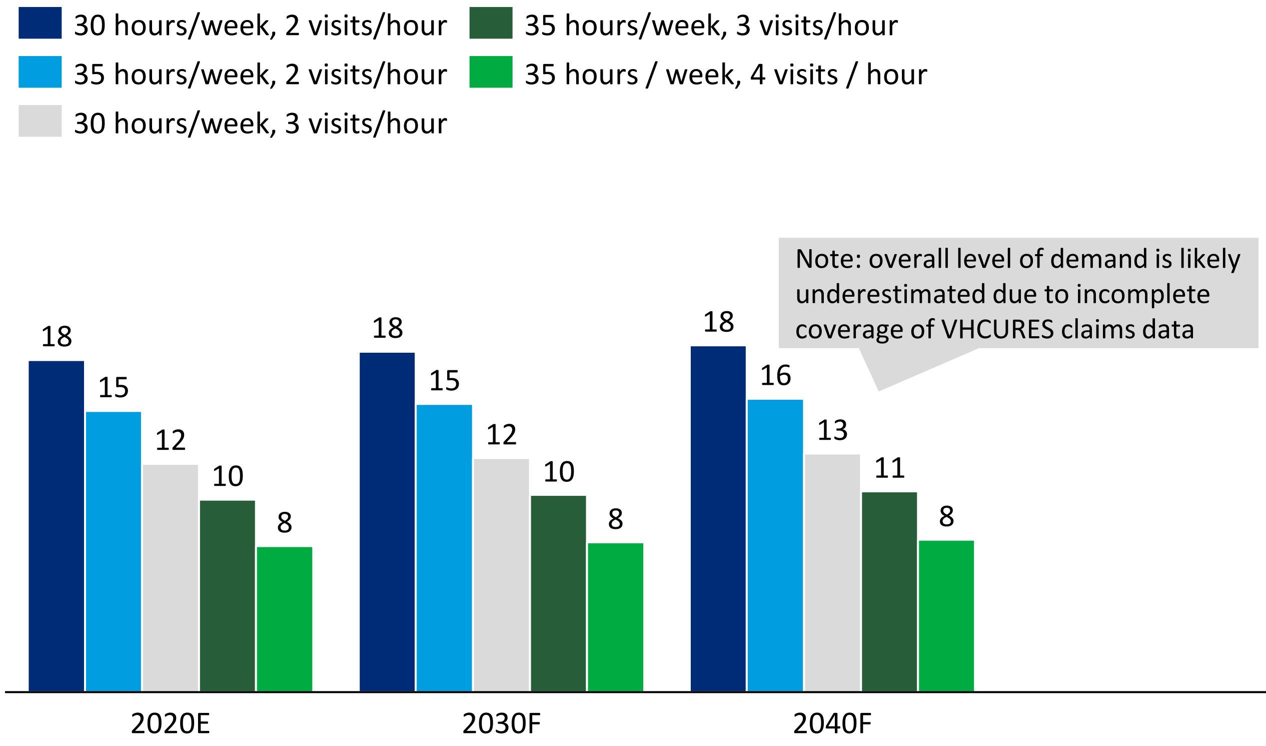
- Based on NIH-referenced study of 392 videotapes of primary care visits between 1998 and 2000 and patient and physician surveys, the average length of visits was **17.4 MINUTES**. The median length of visits was **15.7 MINUTES**[2]
- Based on a cross-sectional study using 4.4MN primary care EMR in 2017, median physician in the sample spent a mean of **18.9 MINUTES** with each patient[3]

1. Assuming 40hr working week, 48 working weeks in a year

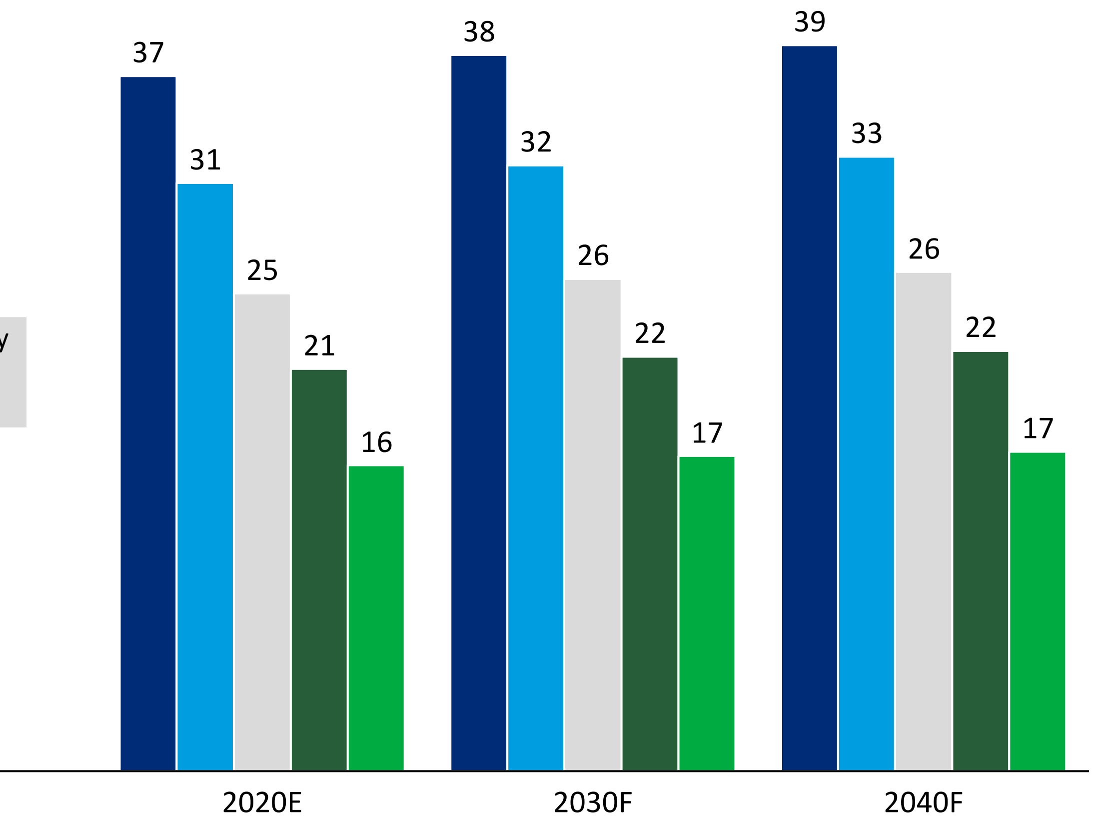
Source: 1. Oliver Wyman Analysis, 2. Time Allocation in Primary Care Office Visits (link), 3. Association of Primary Care Visit Length With Potentially Inappropriate Prescribing (Neprash et al., 2023, link)

IMPROVING CLINICAL VISIT EFFICIENCY AND REDUCING NON-CLINICAL HOURS CAN SIGNIFICANTLY REDUCE PRIMARY CARE FTES REQUIRED

Primary care physician FTE forecast based on claims data[1,2]
2020E-2040F



Primary care physician FTE forecast based on national health statistics[1,3]
2020E-2040F



1. Assuming 40hr working week, 48 working weeks in a year, 1 hour per day for seeing 1 new patient
Source: 2. Commercial, Medicaid and Medicare FFS professional and outpatient claims from VHCURES in calendar years 2021-2022 (note: Commercial claims from VHCURES do not include claims to the Federal Employee Health Benefit Plan, TRICARE (formerly known as Champus), some self-insured commercial plans, plans with a commercial insurer that covers very few Vermont residents, and Veteran Affairs). 3. National health statistics on PCP visits (link), MPR VT Population by HSA, Oliver Wyman analysis

BY 2019/2022 CENSUS, ST. JOHNSBURY HSA HAS 17.0 PRIMARY CARE FTES AND 17.0 APRN FTES, ALLOWING FOR AT LEAST 89,800 PRIMARY CARE VISITS BY RESIDENTS ANNUALLY

Current healthcare workforce in St. Johnsbury HSA

2022 **VT DoH Physician Census**

- **17.0** Primary care FTE by **23** PCPs (**14** FP, **2** IM, **3** OB, **5** PED)
- **1,639** people per PCP FTE (state average **1,593**)
  - Panel size for capitated care is approx. **2000-2200**/MD FTE
- Total PCP patient care hours/week = **695** [695/17.0= **40.9** hrs/FTE]

2019 **VT DOH APRN Census**

- **17.0** Primary Care FTE by **22** APRNs (**3** Adult, **10** FP, **3** Peds, **5** Women's Health, **1** Gerontology)
- **1,672** people per PCP APRN (state average **2,530**)
  - Panel Size for capitated care is approx. **1200-1500**/APRN FTE
- Total PCP APNR patient care hours/week= **789** [789/17= **46.2** hours/FTE]

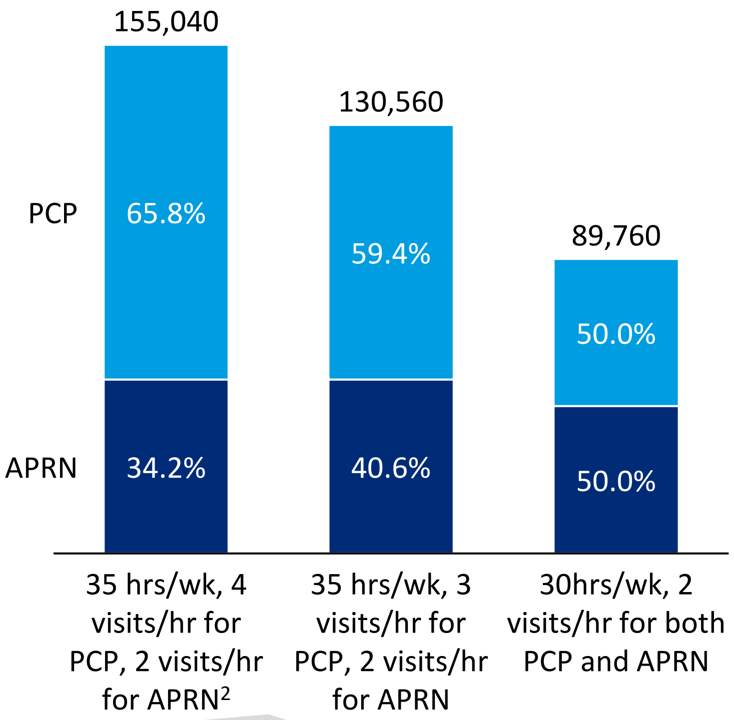
2024 **FQHC data[3]**

Site	MDs	APPs	Total
Concord	0.68	1.38	2.06
Danville	2.8	1.8	4.6
St. Johnsbury	0.8	3.7	4.5

NVRH has a pharmacist embedded in two adult primary care practices.

Estimated annual primary care visit capacity[1,2]

Total number of visits annually by 2019/2022 census

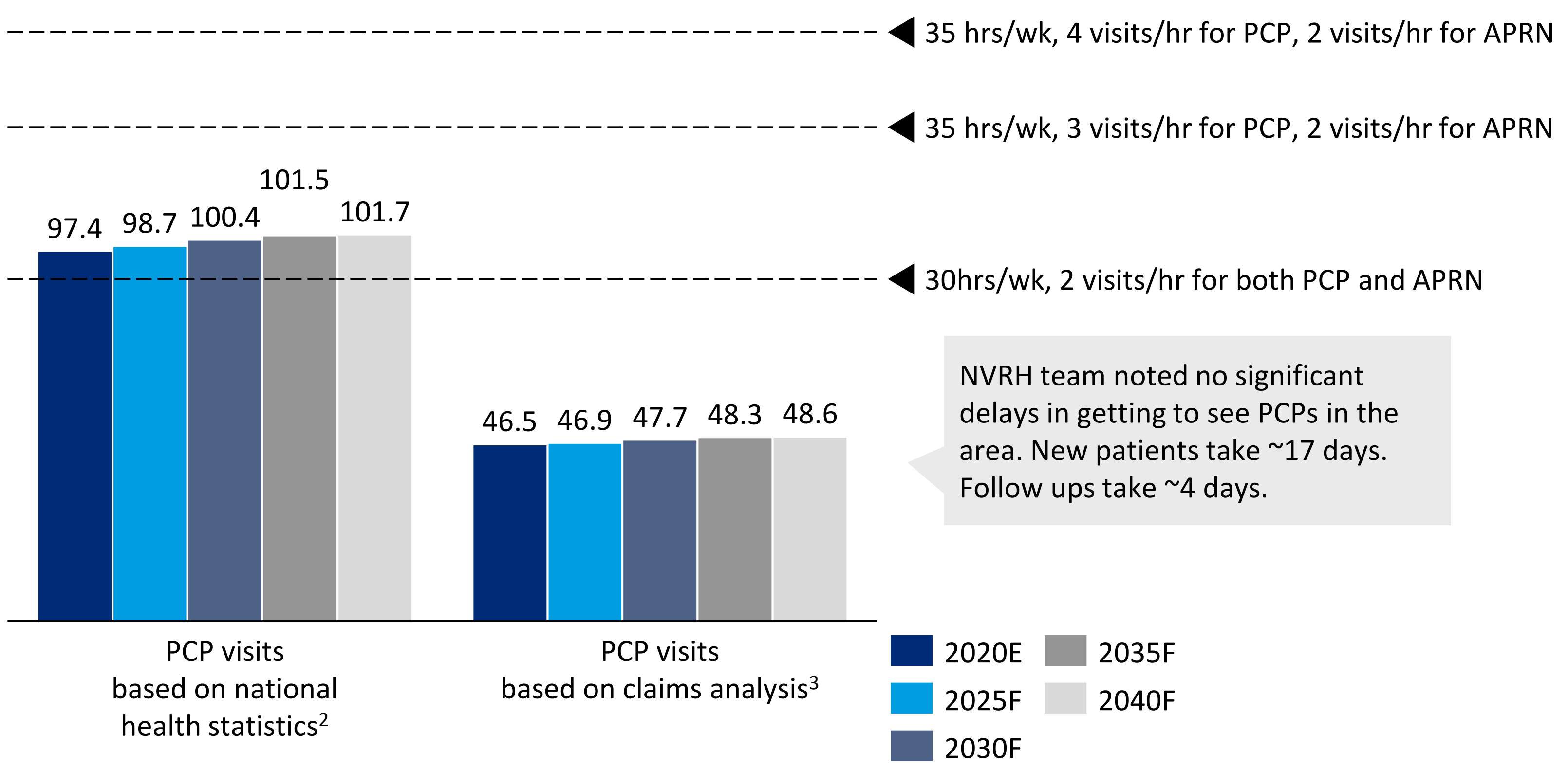


Note: current and future level of primary care FTEs may not sustain 2019/22 level, depending on HCP age and retirement, payer mix and PCP financial viability

Note: 1. Based on 2022 Physician Census and 2019 APRN Census, 2. Assuming 48 working weeks in a year, 1 hour per day for seeing 1 new patient, hours per week are clinical hours excluding admin hours 3. Northern Counties Healthcare FQHC data 2024
Source: 1. 2022 Vermont Department of Health Physician Census Statistical Report (Table 30, 32, link); 2. 2019 Vermont Department of Health Advanced Practice Registered Nurses (APRNs) Census Statistical Report (Table 16,18, link). More Tethered to the EHR: EHR Workload Trends Among Academic Primary Care Physicians, 2019-2023 (Arndt et al, 2024), 2024 FQHC data provided via hospital feedback, Oliver Wyman insights and analysis

CURRENT NUMBER OF PRIMARY CARE FTES ARE LIKELY SUFFICIENT, ASSUMING 2019/20 CENSUS REFLECTS CURRENT FTES AND PRODUCTIVITY CAN BE IMPROVED

Forecasted demand in PCP visits in St. Johnsbury based on different methodology[1]
In thousands, 2020E-2040F



Remarks

- Assuming 2019/2022 PCP and APRN capacity, there should be sufficient FTEs to meet primary care needs, especially if physician productivity is higher than 30 clinical hours per week and 2 visits per hour
  - Given Medicare reimbursement for a 20-min PCP visit is ~$50 per visit and physician salary is $125 per hour, PCPs need to see at least 3 patients per hour to be financially sustainable
- Should current PCP capacity fall below 2019/22 census level, APRNs and pharmacists could be leveraged to take on more primary care responsibilities (e.g. prescribe medications for chronic conditions, administering vaccination)
- Better EMR implementation and care coordination could also reduce admin burden and release more PCP time to clinical work

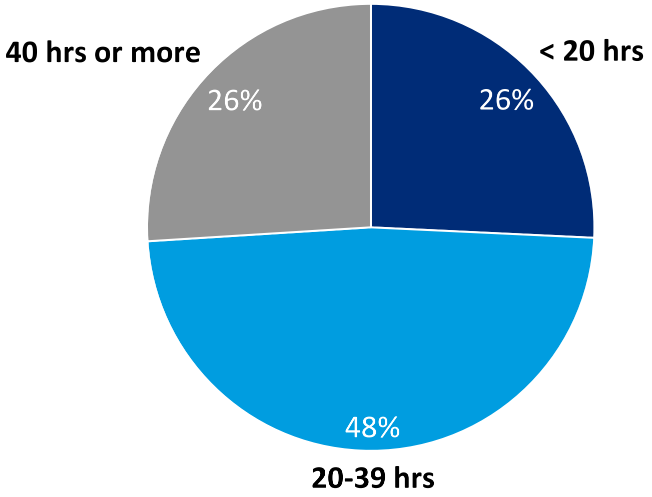
1. Assuming 40hr working week, 48 working weeks in a year, 1 hour per day for seeing 1 new patient
Source: 2. National health statistics on PCP visits (link), MPR VT Population by HSA, Oliver Wyman analysis 3. Commercial, Medicaid and Medicare FFS professional and outpatient claims from VHCURES in calendar years 2021-2022 (note: Commercial claims from VHCURES do not include claims to the Federal Employee Health Benefit Plan, TRICARE (formerly known as Champus), some self-insured commercial plans, plans with a commercial insurer that covers very few Vermont residents, and Veteran Affairs). MPR VT Population by HSA

BY 2021 CENSUS, CALEDONIA COUNTY HAS 95 MENTAL HEALTH COUNSELOR FTES PER 100,000 RESIDENTS

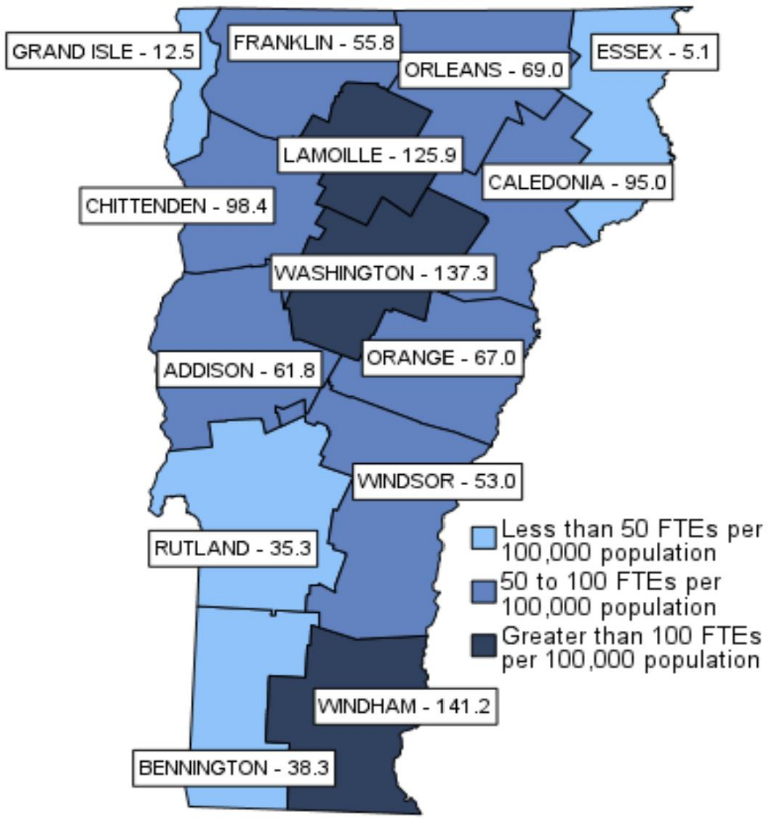
2021 Mental Health Counselor Census

- **547.9** total FTEs were provided by **804** mental health counselors in Vermont
  - 175.5 FTEs, or **32.0%** of total FTEs, were age 60 and older (268 individuals)
- **95** Mental Health Counselor FTEs per 100k population in Caledonia County

Mental Health Counselor FTEs by average weekly hour



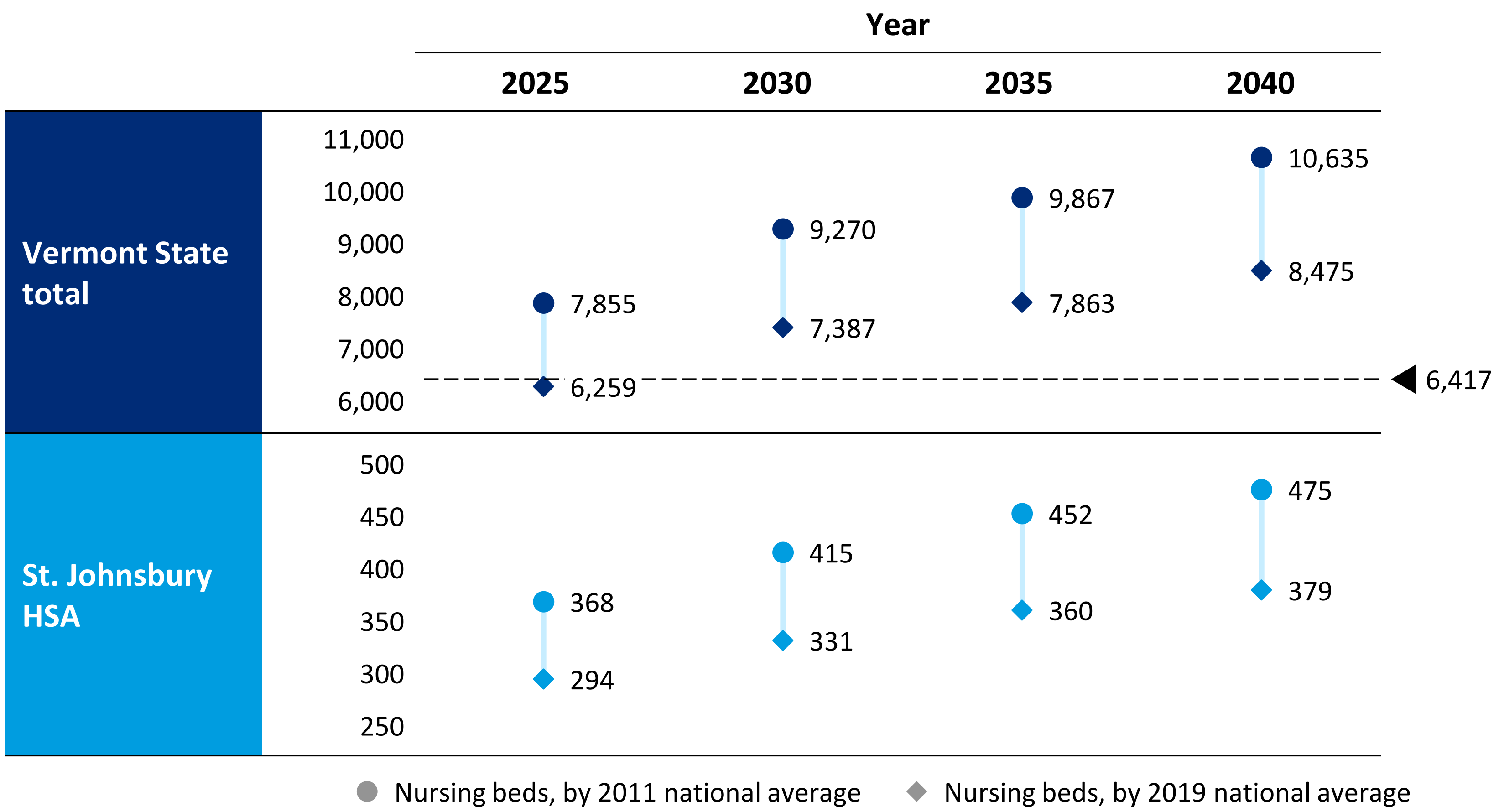
Mental Health Counselor FTEs per 100,000 population by Country



Source: 2021 Health Care Workforce Census Mental Health Counselors (link)

AS POPULATION AGES, VERMONT NEEDS MORE NURSING BEDS TO FULFILL LONG-TERM CARE NEEDS IN THE COMMUNITY AND REDUCE HOSPITAL BOARDERS

Estimated nursing beds needed based on 65+ yrs population projection[2,3]
2025F-2040F



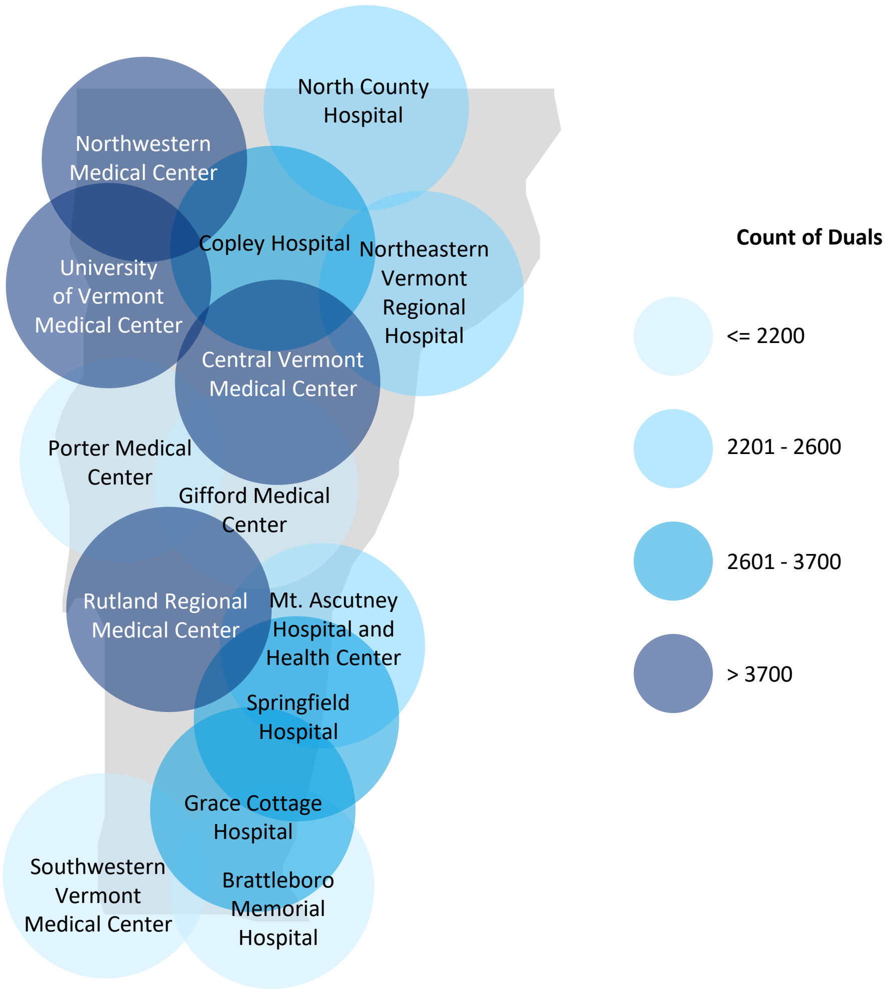
Remarks

- As of January 2020, there are 167 long-term care homes in Vermont. This number includes Nursing Homes (NHs), Assisted Living Residences (ALRs) and Residential Care Homes (RCHs). Together, these homes contain 6,417 licensed beds[3]
- However, capacity has significantly reduced since COVID and persisting nurse staffing challenges
- Based on population projection[1] and US national average of nursing beds per 10,000 65+ yr population[2], by 2040
  - Vermont needs ~8500-10600 nursing beds – an additional 2000-4000 beds compared to current licensed capacity
  - St. Johnsbury HSA needs ~375-475 nursing beds by 2040
  - Choices for Care may reduce these numbers

Source: 1. MPR VT population projection by HSA, 2. Trends in Supply of Nursing Home Beds, 2011-2019 (Miller et al., link), 3. AHS Consumer Guide to VT Long-term Care Facilities (Jan 2020, link)

THERE IS A NEED TO BETTER SUPPORT HIGH UTILIZING DUAL-ELIGIBLE BENEFICIARIES

Count of dual eligibles by 20 mile radius of a hospital, Vermont 2022



Count of Dually Eligible Individuals With a ZIP Code Centroid Within 20 Miles of a Hospital, by Hospital and ZIP Code, 2022

Hospital	ZIP Code	Dual Count
Northeastern Vermont Regional Hospital	05042	24
Northeastern Vermont Regional Hospital	05046	75
Northeastern Vermont Regional Hospital	05050	12
Northeastern Vermont Regional Hospital	05069	42
Northeastern Vermont Regional Hospital	05647	67
Northeastern Vermont Regional Hospital	05658	62
Northeastern Vermont Regional Hospital	05681	25
Northeastern Vermont Regional Hospital	05819	607
Northeastern Vermont Regional Hospital	05821	54
Northeastern Vermont Regional Hospital	05824	67
Northeastern Vermont Regional Hospital	05828	88
Northeastern Vermont Regional Hospital	05832	42
Northeastern Vermont Regional Hospital	05836	65
Northeastern Vermont Regional Hospital	05837	22
Northeastern Vermont Regional Hospital	05838	<11
Northeastern Vermont Regional Hospital	05839	62
Northeastern Vermont Regional Hospital	05840	<11
Northeastern Vermont Regional Hospital	05841	30
Northeastern Vermont Regional Hospital	05842	47
Northeastern Vermont Regional Hospital	05843	223
Northeastern Vermont Regional Hospital	05848	<11
Northeastern Vermont Regional Hospital	05848	<11
Northeastern Vermont Regional Hospital	05849	21
Northeastern Vermont Regional Hospital	05850	40
Northeastern Vermont Regional Hospital	05851	367
Northeastern Vermont Regional Hospital	05858	34
Northeastern Vermont Regional Hospital	05861	<11
Northeastern Vermont Regional Hospital	05862	14
Northeastern Vermont Regional Hospital	05863	13
Northeastern Vermont Regional Hospital	05866	45
Northeastern Vermont Regional Hospital	05867	33
Northeastern Vermont Regional Hospital	05871	117
Northeastern Vermont Regional Hospital	05873	27
Northeastern Vermont Regional Hospital	05904	39
Northeastern Vermont Regional Hospital	05906	86

Source: MPR analysis of dual eligibles using VHCURES 2022 data

VERMONT HAS EXPERIENCED A SIGNIFICANT INCREASE IN HOMELESSNESS SINCE 2007 AND AFTER COVID

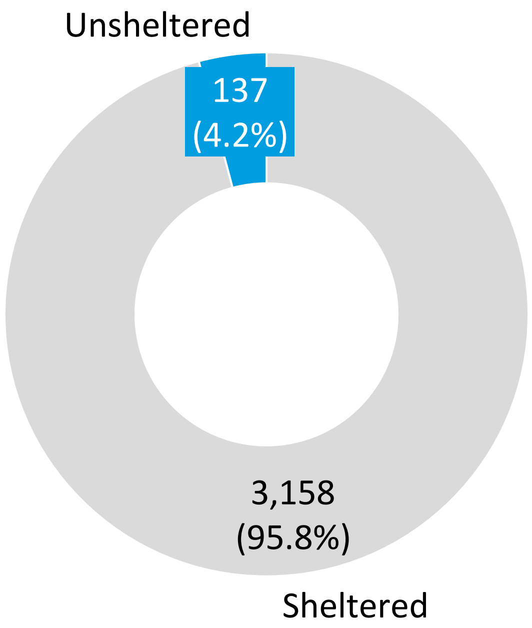
VERMONT

Total homeless, 2023
3,295

18.5% change from 2022

218.4% change from 2007

51 in every **10,000** people
were experiencing sheltered homelessness



Estimates of homelessness

2,129 individuals

1,166 people in families with children

164 unaccompanied homeless youth

122 veterans

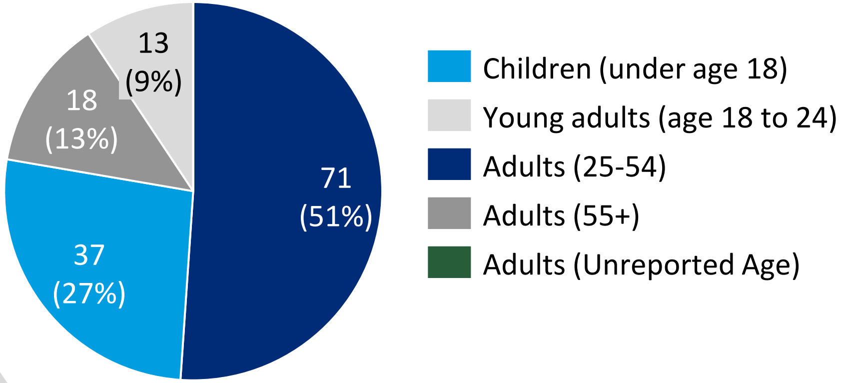
249 chronically homeless individuals

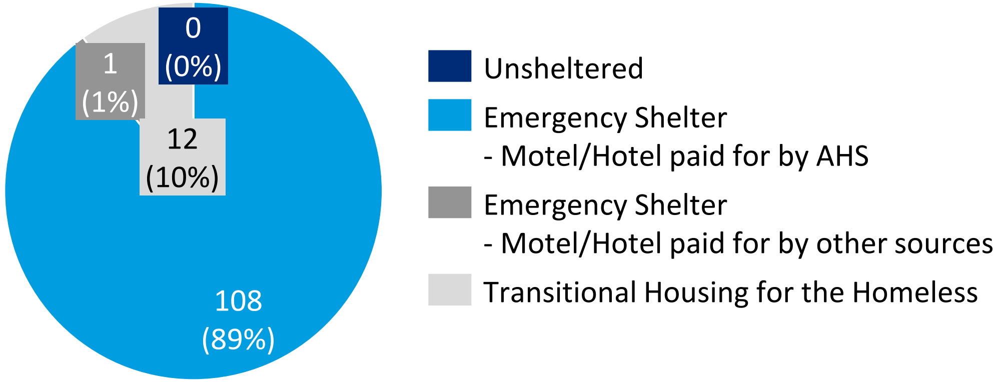
Source: 2023 Annual Homelessness Assessment Report (AHAR) to Congress, Part 1 Point-In-Time Estimates of Homelessness (Dec'23, link)

CALEDONIA COUNTY SAW ~140 VERMONTERS EXPERIENCING HOMELESSNESS IN A POINT-IN-TIME COUNT IN 2022. ~40 HAD A PHYSICAL DISABILITY OR MENTAL HEALTH CONDITION

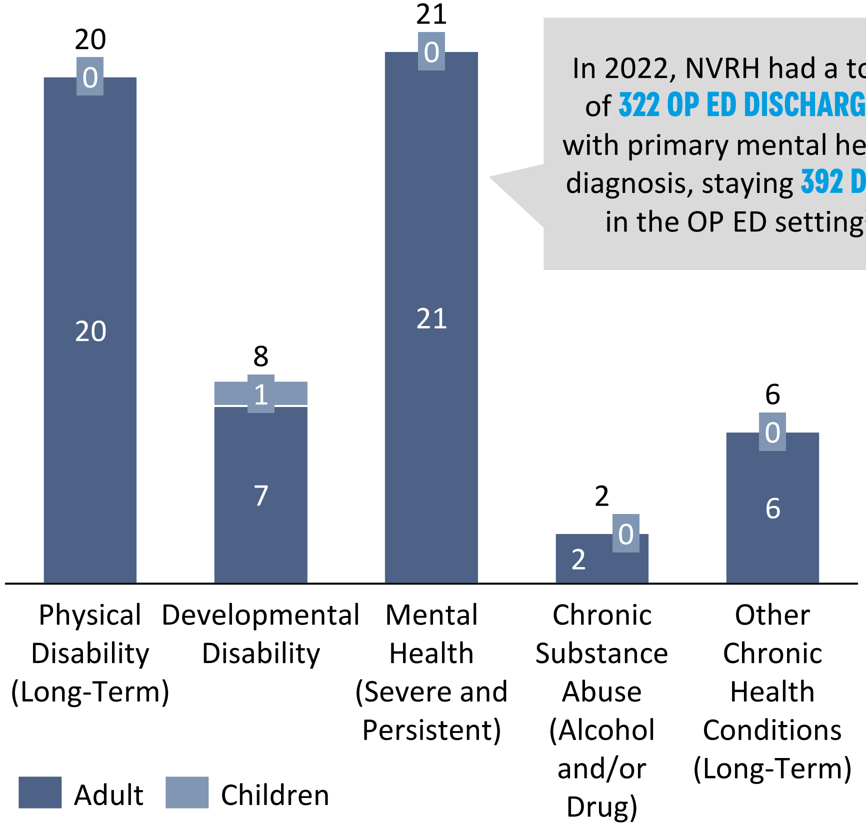
Vermonters experiencing homelessness by age and shelter status
Caledonia County, Point-in-time count on Jan 26, 2022[1]

139
Total number of Vermonters experiencing homelessness in Caledonia County





Vermonters experiencing homelessness with chronic health conditions
Caledonia County, Point-in-time count on Jan 26, 2022[1]



In 2022, NVRH had a total of **322 OP ED DISCHARGES** with primary mental health diagnosis, staying **392 DAYS** in the OP ED setting[2]

Source: 1. Vermont's Annual Point-in-Time Count of sheltered and unsheltered people experiencing homelessness on a single night took place on January 26, 2022 (link), 2. VAHHS analysis using 2022 hospital discharge data

ST. JOHNSBURY

Individual quotes from HSA-level community calls held Aug-Nov 2023

EXPERIENCES SHARED (ANONYMIZED QUOTES)

RECOMMENDED ACTIONS (PRELIMINARY)

I. Gaps in culturally competent care

It's hard to find a doctor, it's a 50 min drive to find my doctor. It's not hard getting an appointment but I need to drive to NH for 3hrs to get treatment

Due to my insurance, I have very limited access to mental health services

For 2 years during COVID, my mom didn't receive an email or phone call asking her how she was. This isolation can be very detrimental

Part of the reason is there are not enough doctors, part of the reason is that we have a lot of doctors working part-time

- **Invest in Healthcare Infrastructure:** Invest in infrastructure such as primary care services, mental health facilities, and substance abuse provided when and where all populations can access (Develop strategies to attract more healthcare providers to the area)
- **Improve Provider Diversity:** Increase efforts to recruit and retain healthcare providers from diverse backgrounds and who understand the local culture
- **Improve Insurance Coverage:** Advocate for policies that expand insurance coverage for mental health services
- **Localize Mental Health Services:** Develop local mental health resources and clinics to improve access

II. Gaps in culturally competent and psychologically safe working environment

Mental Health clinicians are leaving for better employment elsewhere. Paperwork is a big admin burden.

As a nurse, I feel disrespected often

Violence in ED in common

- **Respect and Support for Healthcare Workers:** Implement programs to improve workplace culture, such as respect training and mental health support for staff and emphasize need to create a supportive and safe working environment for all
- **Address Workforce Issues:** Continue investments in workforce development, offer competitive salaries, and reduce administrative burdens to retain healthcare professionals

III. Lack of coordination with care givers and community services

I was attending the University and needed to change providers for MH services, this took way too long.

There is misinformation in the process for choosing a Medicare plan

- **Enhance Community Education:** Provide community education on healthcare processes, such as choosing a Medicare plan, in highly accessible ways (e.g., multiple languages, recordings)

VERMONT HEALTH EQUITY HEAT MAP

% of total population of county, unless otherwise stated (darker colors are higher values)

HSA (Primary)	HSA (Secondary)	County	Adults with Disability	65+	Immigrant/ Refugee	Veterans	Farmworkers that are Migrant Workers	Social Vulnerability Index (SVI)	SVI for BIPOC Population	Below Poverty	Experiencing Food Insecurity	Adults with Suicidal Thoughts	Households with Language Isolation	Experiencing Homelessness
State			27%	20%	4.4%	2.6%	10.2%			10.5%	8.9%	6%	0.6%	0.43%
Barre		Washington	25%	20%	2.9%	6.2%	14.0%	0.150	0.166	10.0%	8.5%	5%	1.0%	0.65%
Burlington		Chittenden	21%	15%	8.2%	5.4%	9.4%	0.179	0.379	11.3%	7.9%	6%	1.3%	0.40%
Morrisville		Lamoille	27%	17%	3.0%	6.3%	No data	0.265	0.136	10.5%	8.9%	7%	0.1%	0.22%
Randolph	White River Junction	Orange	27%	21%	1.8%	7.2%	7.4%	0.229	0.077	9.5%	8.0%	5%	0.2%	0.13%
Newport	Morrisville	Orleans	32%	23%	3.7%	7.6%	6.1%	0.384	0.098	10.4%	9.7%	7%	0.6%	0.12%
Newport	St. Johnsbury	Essex	29%	26%	2.3%	10.9%	No data	0.573	0.086	14.7%	11.9%	No data	0.6%	0.02%
St. Johnsbury		Caledonia	30%	21%	2.4%	7.2%	5.5%	0.251	0.145	12.6%	9.7%	8%	0.2%	0.46%
St. Albans		Franklin	28%	16%	2.6%	7.1%	9.8%	0.277	0.166	9.4%	8.2%	5%	0.5%	0.32%
St. Albans	Burlington	Grand Isle	28%	21%	3.3%	8.6%	14.2%	0.170	0.241	6.5%	6.4%	No data	0.2%	0.01%
Middlebury	Burlington	Addison	22%	20%	4.9%	6.0%	14.8%	0.174	0.234	7.0%	7.3%	No data	0.1%	0.26%
Rutland		Rutland	28%	22%	2.2%	7.1%	2.6%	0.221	0.107	10.5%	9.6%	3%	0.3%	0.62%
Bennington		Bennington	30%	23%	3.1%	7.1%	No data	0.285	0.166	11.3%	10.1%	No data	0.4%	0.74%
White River Junction	Springfield	Windsor	28%	24%	3.7%	7.5%	7.1%	0.186	0.118	9.1%	8.2%	7%	0.1%	0.24%
Brattleboro		Windham	30%	23%	3.0%	6.6%	21.9%	0.373	0.224	13.4%	11.0%	7%	0.4%	0.88%

Focus

St. Johnsbury population quick facts (relative to other Vermont HSAs)

- POPULATION DEMOGRAPHICS: Highest veteran population and on higher end of population of adults with disabilities
- KNOWN BARRIERS TO EQUITABLE CARE: Highest population below poverty level, experiencing food insecurity, adults with suicidal thoughts

Source. 1. Behavioral Risk Factor Surveillance System 2022 Report (published Jan 2024) 2. Data.Census.Gov 3.https://stacker.com/vermont/counties-most-veterans-vermont 4. CDC/ATSDR Social Vulnerability Index 5. United States Department of Agriculture/Feeding America 6. Vermont Coalition to end homelessness PIT report 2022

HOSPITAL MISSION & MEDICAL/SERVICE NEEDS IN THE ST. JOHNSBURY HSA

Northeastern Vermont Regional Hospital

Mission

Committed to improving the health and well-being of all

Vision

Providing exceptional care in an environment where our patients, community, and employees thrive

St. Johnsbury Medical & Service Needs

- **More accessible** transportation for people to and from urgent/acute medical visits and home from the ED/hospital
- **More timely** transportation for patients between healthcare institutions
- **Housing** for the unhoused and those with limited means
- **Pharmacy services** and **rapid drug delivery**
- **Timely access** to an appropriate level of mental health treatment
- Access to **needed specialty services without the need to travel** a great distance (DHMC or UVM)
- **Local** access to SNF and extended care facilities

Source: Hospital community health needs assessments, OW notes from Phase 1 discussion with hospital leadership team and board members

IN THE HOSPITAL VIEW, WE WILL DISCUSS

Current hospital financials

- 2018-2023 financials
- Future projections

Potential levers to improve care quality and hospital sustainability

- Avoidable ED and IP
- Avoidable boarding days
- Current low volume services
- Potential services to be recaptured

Recommendations

- Hospital-level recommendations and options
- State-level enablers
- Health equity recommendations

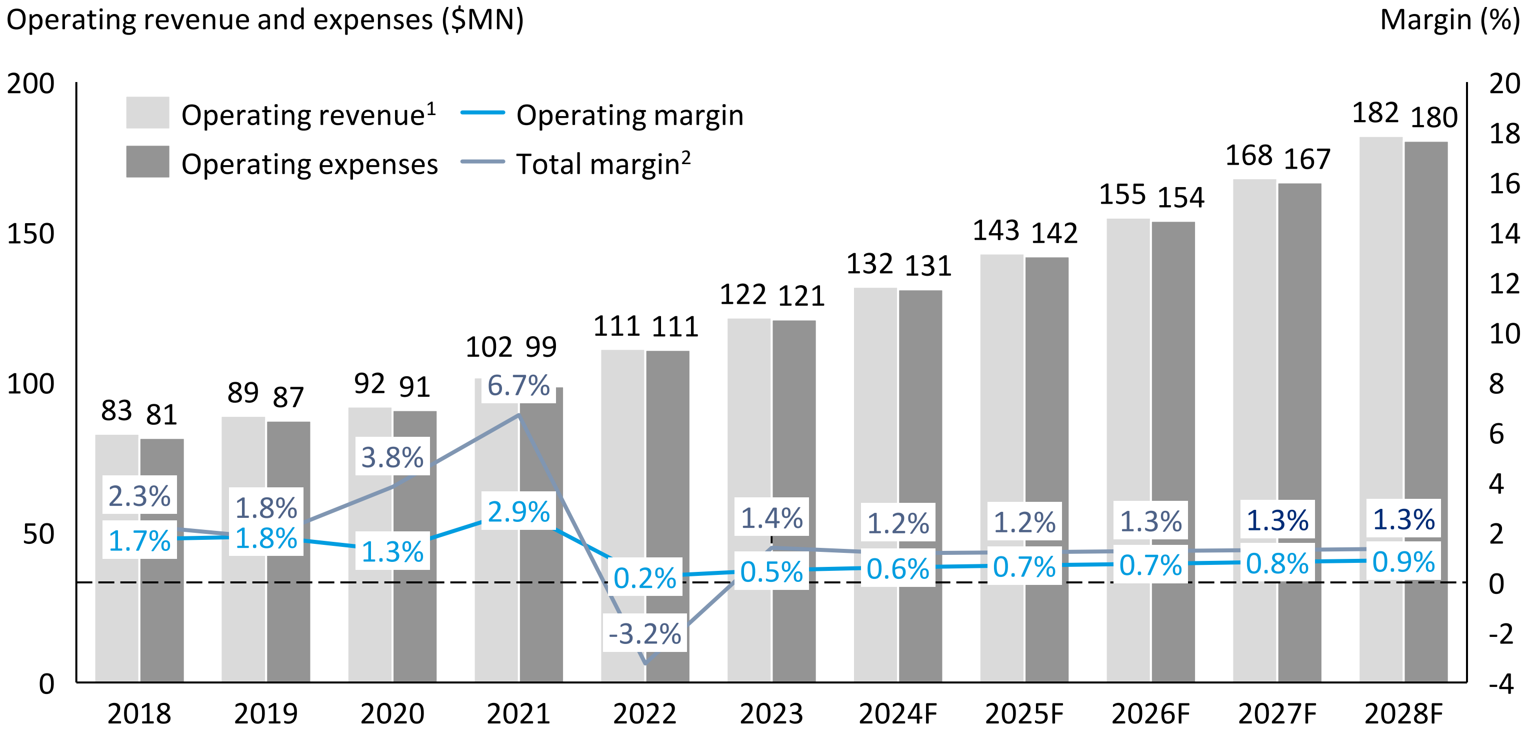
4.1

REVENUE AND EXPENSE TRENDS

IF HISTORICAL REVENUE AND EXPENSE TRENDS CONTINUE, NVRH COULD MAINTAIN A LOW POSITIVE MARGIN, BUT SUBJECT TO HIGH FINANCIAL RISK UPON VOLATILITY

Northeastern Vermont Regional Hospital (NVRH) operating revenue, expenses and margin
$MN, %, 2018-2028F

Projected total operating revenue growth $CAGR_{23-28}$: **8.4%**; Projected total operating expense growth $CAGR_{23-28}$: **8.3%**



Remarks

- Between 2018 and 2023, NVRH's operating expenses grew at 8.2% $CAGR_{18-23}$, consistently outpacing operating revenue[1] growth at 8.0% $CAGR_{18-23}$
- Average total margin between 2018 and 2023 is 2.1%, but with high volatility driven by income and losses from investments and non-operating revenue
- If the current revenue and expense growth trends were to continue, NVRH could maintain a low positive margin, but would be subject to high financial risk due to non-operating revenue volatility

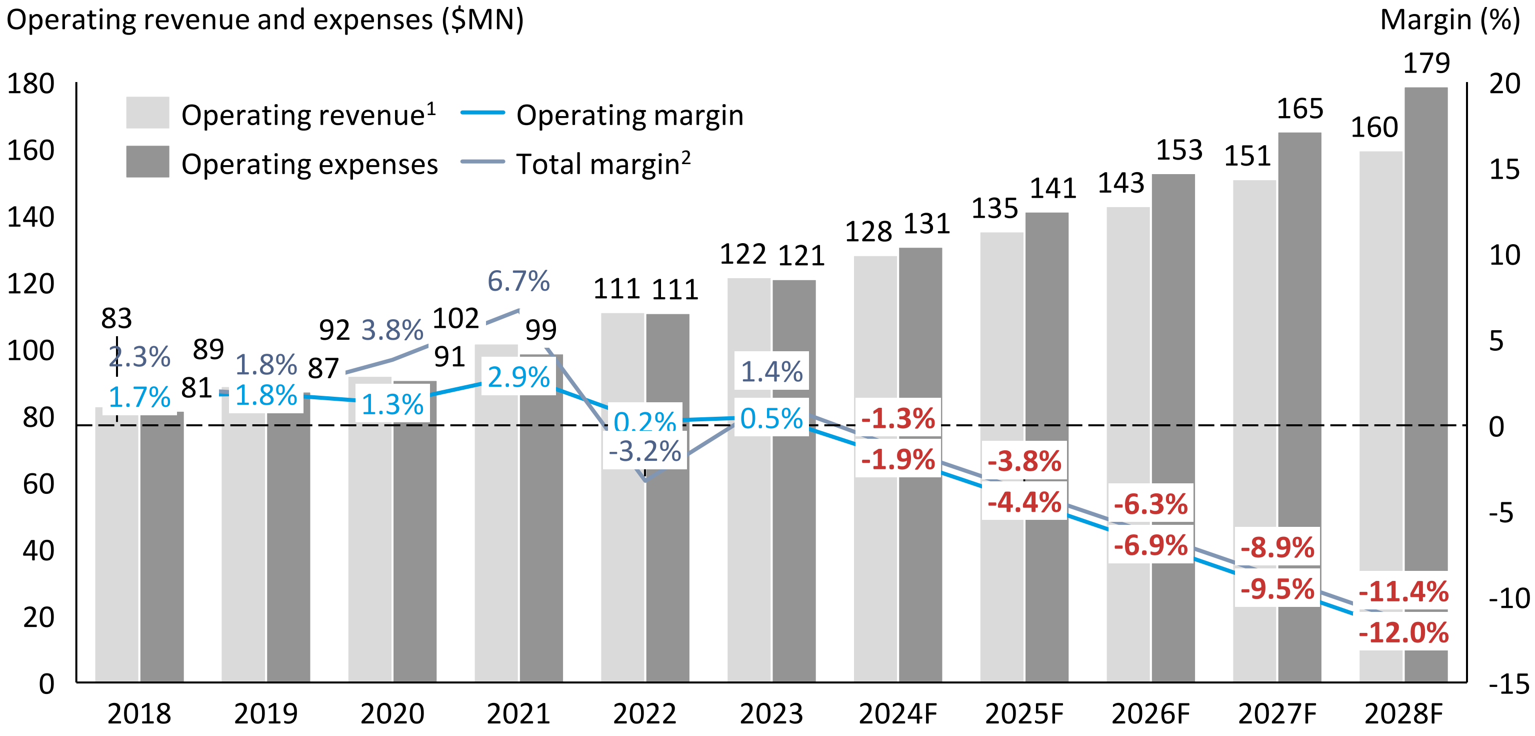
If Medicare Advantage and Medicaid revenue only grow at 3.5%, **Commercial revenue** would need to increase annually by **14.1%** to ensure positive operating margin by 2028 or **15.4%** to get to 3% operating margin by 2028[4]

1. Operating revenue include Covid-19 related government supports in 2020-22; 2. Total margin include income (losses) from investments and nonoperating revenue; 3. Assuming forecasted non-operating income is at the average level between 2018 and 2023, with 3% CAGR in line with inflation; 4. Assuming 2023 non-340B revenue and non-tax/depreciation/amortization/interest expense payer mix at 40% commercial, 41% Medicare FFS, 5% Medicare Advantage, 13% Medicaid, and 1% uncategorized payers; revenues from Medicare Advantage, Medicaid and uncategorized payers grown at 3.5% annually from 2024 onwards; 5. Assuming projected tax rate is 5.6% of net patient revenue; 6. Assuming Medicare FFS reimburses at 101% cost; 7. Assuming 340B payments remain constant, with growth due to inflation only. 6. Average approved commercial rate 2018-2023 for NVRH was 4.3%. Average approved commercial rate 2021-2023 was 5.9% Source: GMCB hospital financial records, Oliver Wyman analysis.

ASSUMING 3.5% ANNUAL GROWTH ON NON-340B REVENUE AND ~8% GROWTH IN EXPENSES, NVRH COULD START TO BECOME FINANCIALLY NONVIABLE IN 2024

Northeastern Vermont Regional Hospital (NVRH) operating revenue, expenses and margin
$MN, %, 2018-2028F

Projected total operating revenue growth $CAGR_{23\text{-}28}$: **5.6%**; Projected total operating expense growth $CAGR_{23\text{-}28}$: **8.1%**



Remarks

- If growth in expense continuously outpace growth in revenue, NVRH could start to become financially nonviable as early as 2024

Annual growth assumptions from 2023 to 2028	
Operating revenue	
340B revenue	3% (inflation only)
Non-340B revenue (except Medicare FFS)	3.5%
Non-operating revenue	3% vs. average revenue in 2018-23
Operating expenses	
Physician salary & benefits	5%
Non-MD salary & benefits	10%
Other operating expense	7%
Depreciation / amortization	$CAGR_{18\text{-}23}$ (i.e. 7%)
Interest	$CAGR_{18\text{-}23}$ (i.e. 9.5%)

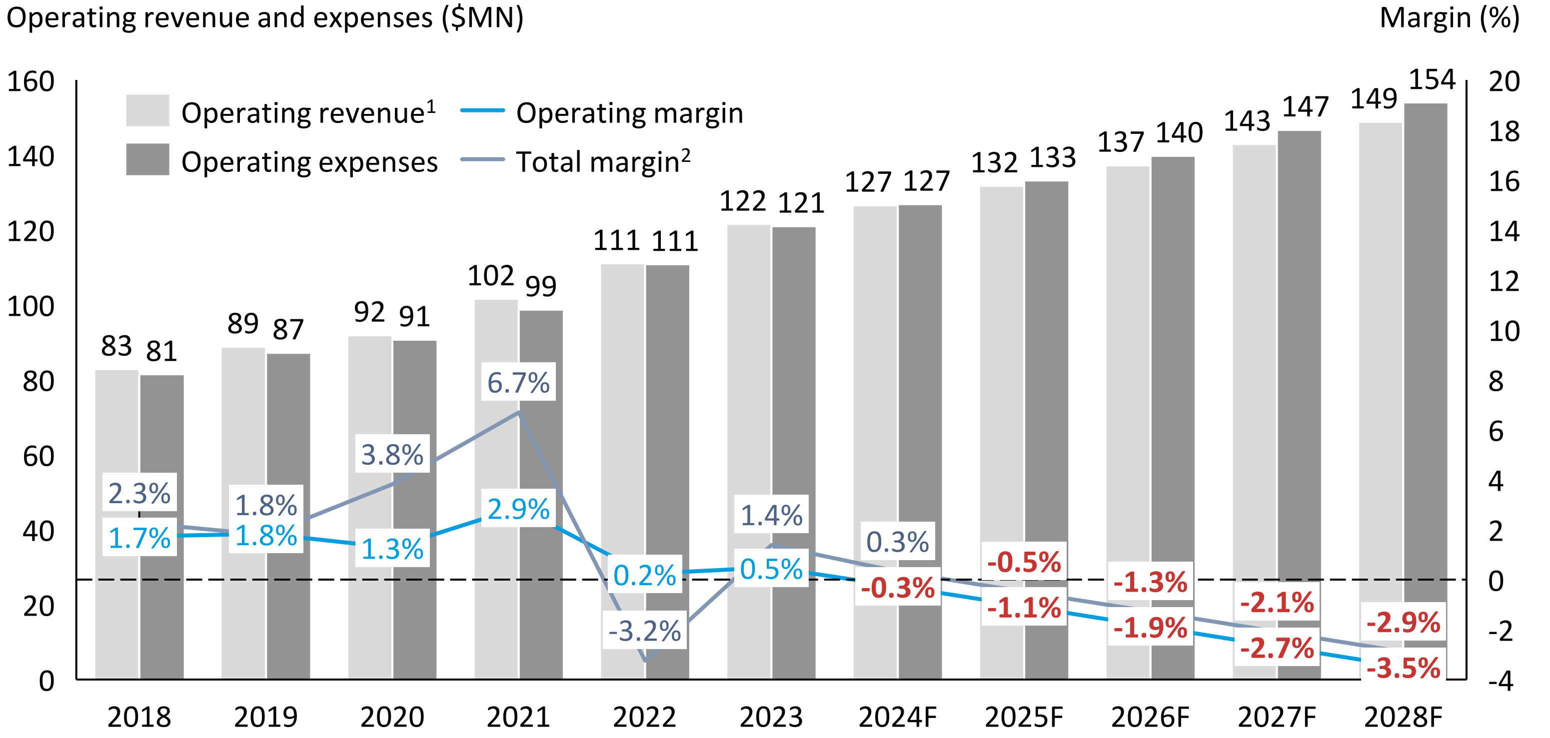
1. Operating revenue include Covid-19 related government supports in 2020-22; 2. Total margin include income (losses) from investments and nonoperating revenue; 3. Assuming forecasted non-operating income is at the average level between 2018 and 2023, with 3% CAGR in line with inflation; 4. Assuming 2023 non-340B revenue and non-tax/depreciation/amortization/interest expense payer mix at 40% commercial, 41% Medicare FFS, 5% Medicare Advantage, 13% Medicaid, and 1% uncategorized payers; revenues from Medicare Advantage, Medicaid and uncategorized payers grown at 3.5% annually from 2024 onwards; 5. Assuming projected tax rate is 5.6% of net patient revenue; 6. Assuming Medicare FFS reimburses at 101% cost; 7. Assuming 340B payments remain constant, with growth due to inflation only. Source: GMCB hospital financial records, Oliver Wyman analysis

ASSUMING 3.5% ANNUAL GROWTH ON NON-340B REVENUE AND ~5% GROWTH IN TOTAL OPERATING EXPENSES, NVRH WOULD STILL SEE CONSISTENT NEGATIVE MARGINS

Northeastern Vermont Regional Hospital (NVRH) operating revenue, expenses and margin

$MN, %, 2018-2028F

Projected total operating revenue growth $CAGR_{23-28}$: **4.1%**; Projected total operating expense growth $CAGR_{23-28}$: **4.9%**



Remarks

- If growth in expense continuously outpace growth in revenue, NVRH could start to become financially nonviable as early as 2025

Annual growth assumptions from 2023 to 2028	
Operating revenue	
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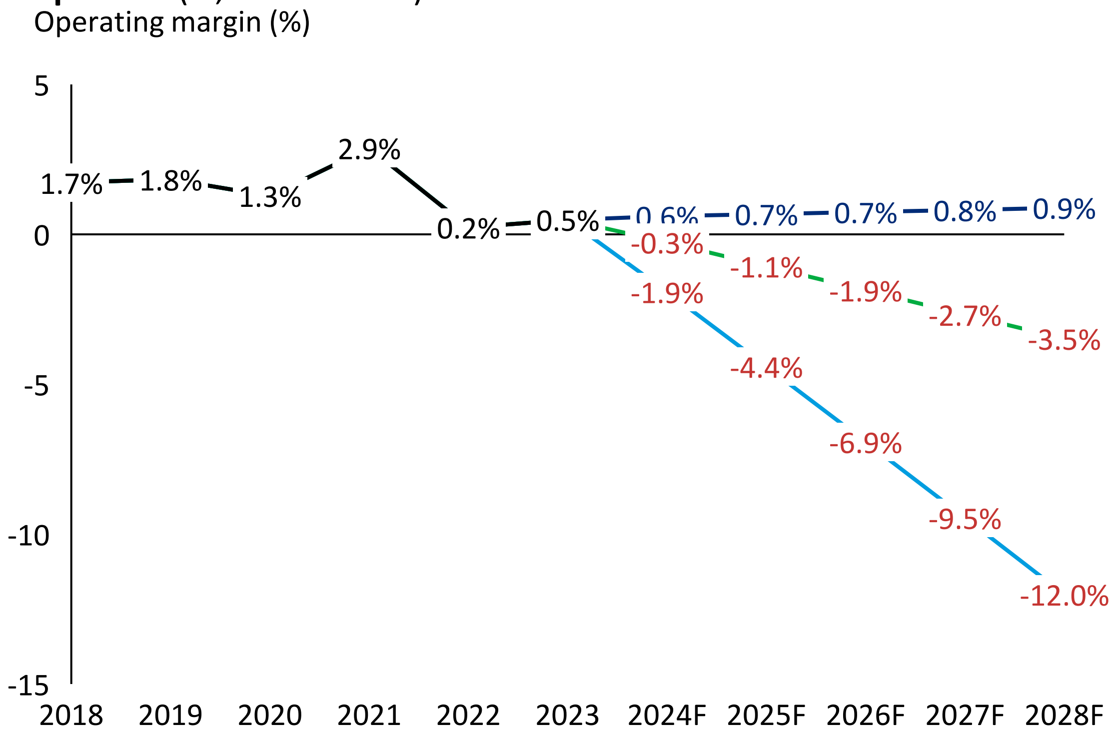
Non-340B revenue (except Medicare FFS)	3.5%
Non-operating revenue	3% vs. average revenue in 2018-23
Operating expenses	
Physician salary & benefits	5%
Non-MD salary & benefits	5%
Other operating expense	5%
Depreciation / amortization	5%
Interest	5%

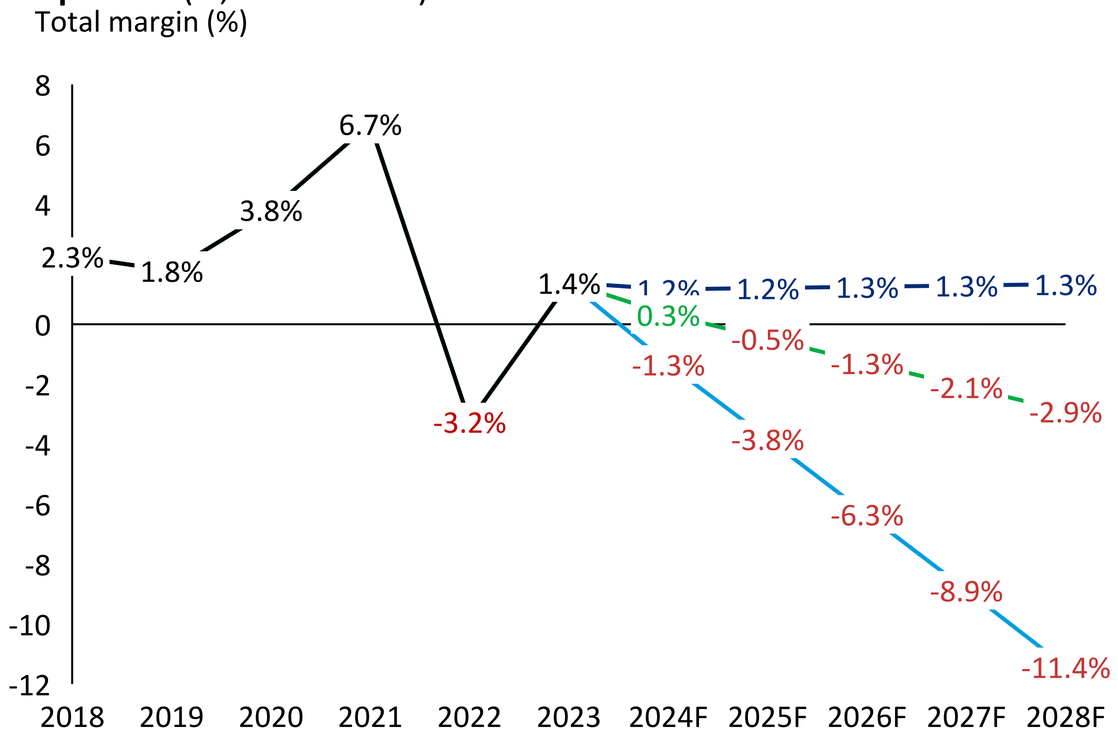
1. Operating revenue include Covid-19 related government supports in 2020-22; 2. Total margin include income (losses) from investments and nonoperating revenue; 3. Assuming forecasted non-operating income is at the average level between 2018 and 2023, with 3% CAGR in line with inflation; 4. Assuming 2023 non-340B revenue and non-tax/depreciation/amortization/interest expense payer mix at 40% commercial, 41% Medicare FFS, 5% Medicare Advantage, 13% Medicaid, and 1% uncategorized payers; revenues from Medicare Advantage, Medicaid and uncategorized payers grown at 3.5% annually from 2024 onwards; 5. Assuming projected tax rate is 5.6% of net patient revenue; 6. Assuming Medicare FFS reimburses at 101% cost; 7. Assuming 340B payments remain constant, with growth due to inflation only. Source: GMCB hospital financial records, Oliver Wyman analysis

REALISTIC ASSUMPTIONS ON FUTURE REVENUE AND EXPENSE GROWTH IMPLY CHALLENGING FINANCIAL PROJECTIONS AHEAD

Northeastern Vermont Regional Hospital <u>operating</u> margin forecast comparison (%, 2018-2028F)



Northeastern Vermont Regional Hospital <u>total</u> margin forecast comparison (%, 2018-2028F)

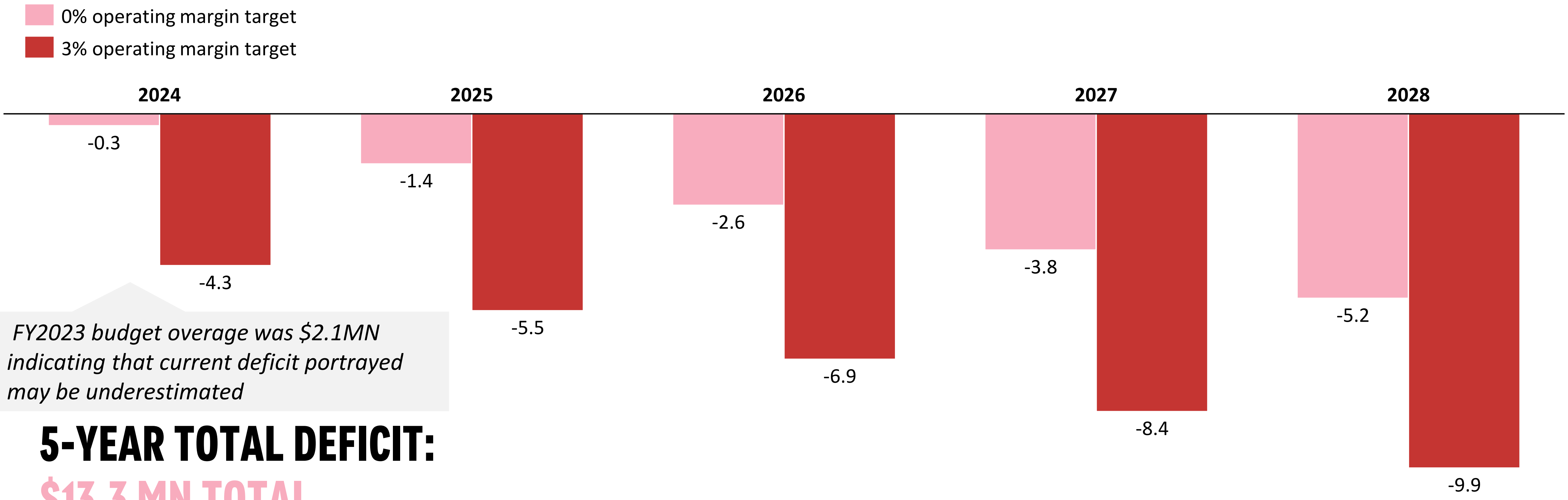


Future revenue and expense growth = CAGR18-23
3.5% non-340B revenue and 5% expense growth assumptions
OW revenue and expense growth assumptions
Historical margin

1. Total margin include income (losses) from investments and nonoperating revenue. 2. See previous slides for specific growth assumptions
Source: GMCB hospital financial records, Oliver Wyman analysis

IF EXPENSES GROW AT 5%, NVRH WILL NEED AN ADDITIONAL ~$13.3 MN OVER THE NEXT FIVE YEARS TO BREAK EVEN AND $35.0 MN TO ACHIEVE 3% OPERATING MARGIN

Financial deficit vs. revenue needed to achieve 0% or 3% operating margin, assuming 3.5% revenue and 5% expense growth[1]
2024F-2028F, USD MN



FY2023 budget overage was $2.1MN indicating that current deficit portrayed may be underestimated

5-YEAR TOTAL DEFICIT:

$13.3 MN TOTAL vs. break-even

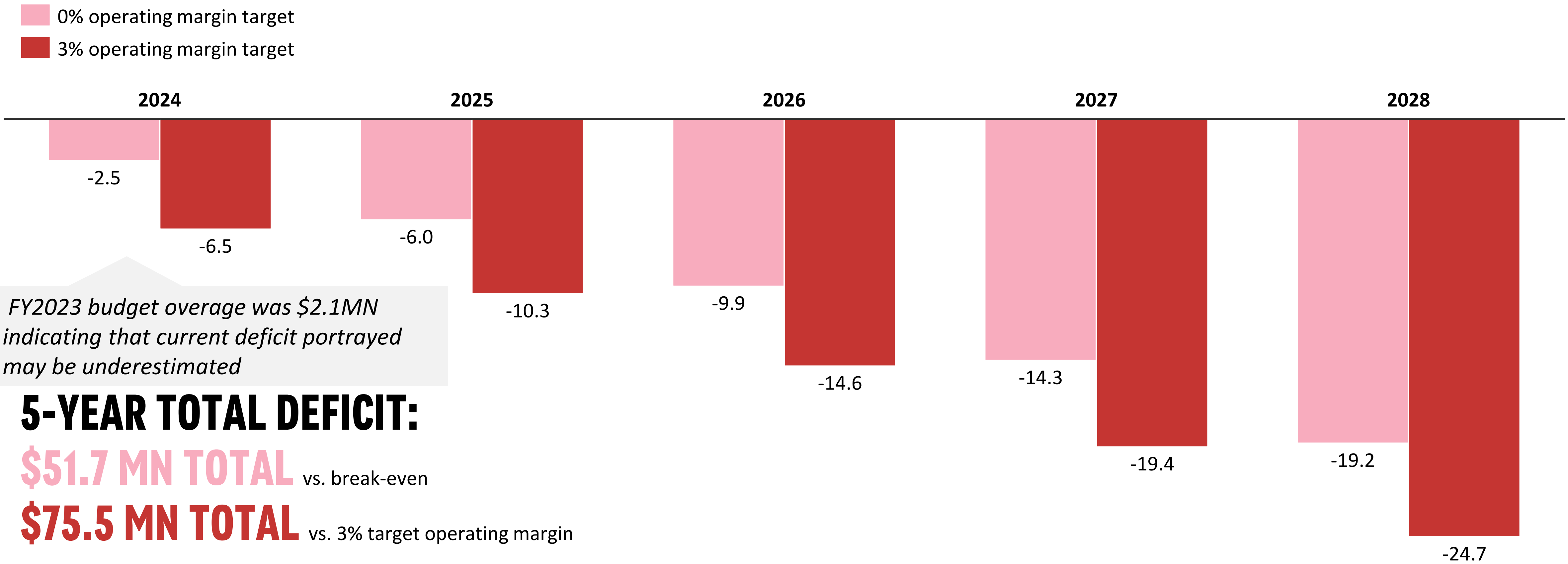
$35.0 MN TOTAL vs. 3% target operating margin

1. Assuming non-340B revenue annual growth rate at 3.5% and 5% operating expense rate growth from 2023 baseline

Source: GMCB hospital financial records, Oliver Wyman analysis

IF EXPENSES GROW AT 7-8%, NVRH WILL NEED AN ADDITIONAL ~$51.7 MN OVER THE NEXT FIVE YEARS TO BREAK EVEN AND $75.5 MN TO ACHIEVE 3% OPERATING MARGIN

Financial deficit vs. revenue needed to achieve 0% or 3% operating margin, assuming 3.5% revenue and 7-8% expense growth[1]
2024F-2028F, USD MN



FY2023 budget overage was $2.1MN indicating that current deficit portrayed may be underestimated

5-YEAR TOTAL DEFICIT:

$51.7 MN TOTAL vs. break-even

$75.5 MN TOTAL vs. 3% target operating margin

1. Assuming 340B payments remain constant, with growth due to inflation only at 3%, 3.5% annual non-340B revenue growth, 10% annual expense growth on non-MD salaries and benefits, 5% annual expense growth on physician fees, salaries, and benefits, and 7% growth in other operating expenses from 2024 onwards

Source: GMCB hospital financial records, Oliver Wyman analysis

4.2

POTENTIALLY AVOIDABLE ED AND IP VOLUMES

ACROSS THE STATE OF VERMONT, KEY UPFRONT CONDITIONS THAT NEED TO BE TRUE FOR ED & IP IMPACT ARE NOT CURRENTLY MET, AND WILL REQUIRE SEVERAL YEARS TO REALIZE

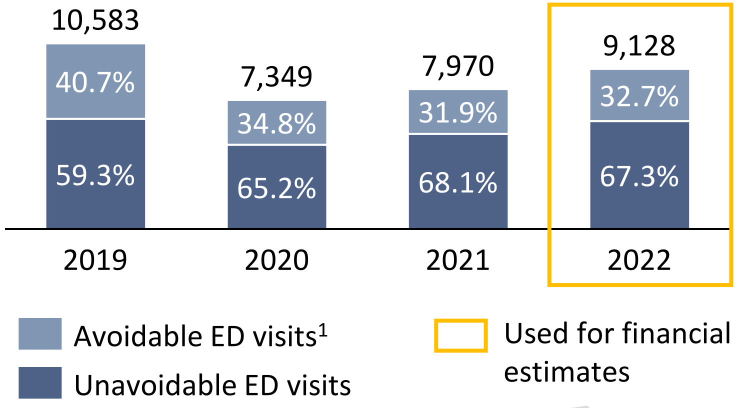
Conditions that need to be true to improve hospital efficiency through reducing avoidable visits	
1	**Adequate** and **timely** (7 days for routine appointment and same day for urgent) access to primary care and preventive services exists
2	**Adequate** and **timely** transportation to medical and other acutely needed appointments within 12 hours of a request.
3	Places **available** for patients discharged from the hospital or ED to receive the **next most appropriate** level of care (homes, Home Health, SNF and mental health)
4	Changes in **care delivery expectations** (work effort and call schedules) and **new types** of providers
5	**Robust** information technology systems - **both state-wide** and **within provider organizations**

In the current state:

- **None** of the following conditions are currently met
- The conditions will require **several years of intensive, focused, sustained effort by all parties** to come to realization

10-30% OF CURRENT ED VISITS AND 20% OF CURRENT IP ADMISSIONS AT NVRH ARE AVOIDABLE

Number of ED visits in NVRH in 2019-22[1,3]
MPR analysis using NYU ED algorithm

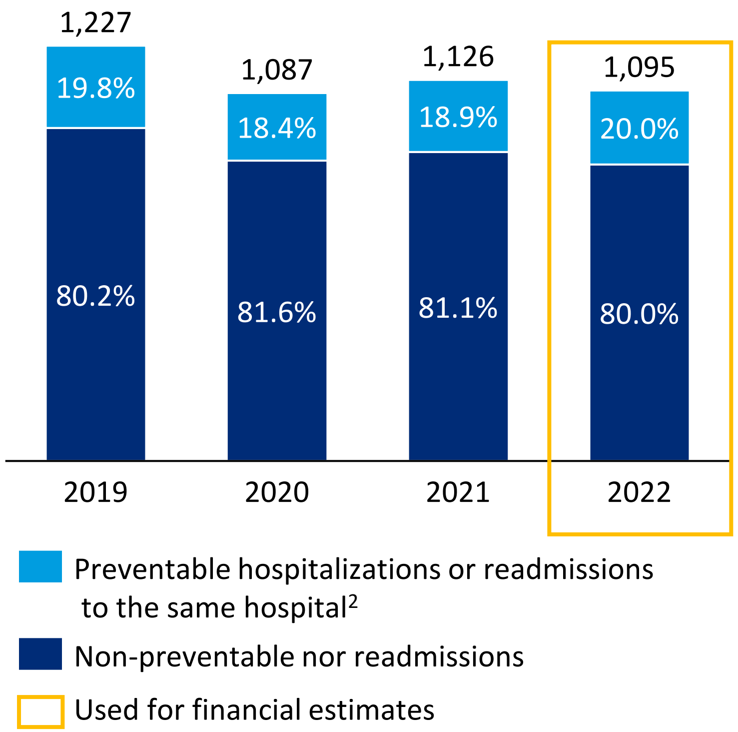


In OW Survey using 2022 VAHHS data, NVRH reported ~34.8% of ED discharges were preventable.

Percentage of avoidable ED visits in 2019-22[4]
NVRH analysis using Blueprint codes

2019	2020	2021	2022
17.2%	11.8%	9.7%	9.6%

Number of acute IP stays in NVRH[2,3]
2019-2022



Remarks

- Hospital ED capacity is consumed when other forms of care are inaccessible locally (e.g. primary care, urgent care, pediatric dental care)
- Improved access to urgent and primary care could result in a significant reduction in ED visits as well as improve health outcomes
- Some IP utilization can be avoided to improve throughput both by improving hospital quality and enhancing community care and resources

1. Avoidable Emergency Department (ED) use refers to ED visits that could have been treated in a primary care setting or avoided through timely and appropriate primary and specialty care. We identified ED visits as avoidable from their principal diagnosis codes, classified using the New York University (NYU) ED algorithm (Billings et al., 2000), updated with a patch accounting for new International Classification of Diseases ICD-10 codes released since the algorithm's original publication (Johnston et al., 2017)
2. Avoidable hospital use refers to preventable hospitalizations and unplanned 30-day readmissions after an inpatient discharge. Preventable hospitalizations were identified using the Prevention Quality Overall Composite Indicator #90 (PQI #90), developed by the Agency for Healthcare Research and Quality. 30-day unplanned readmissions were identified using the HEDIS Plan All-Cause Readmissions (PCR) measure developed by the National Committee for Quality Assurance. The data above aggregate acute inpatient claims into stays by combining overlapping claims or clams indicating transfers to a different hospital for continuing care. Preventable hospitalizations with readmissions were combined to arrive at deduplicated counts for total avoidable hospital use. Since some preventable hospitalizations are also unplanned readmissions and vice versa, duplications were removed to avoid double-counting.
Source: 3. Commercial, Medicaid and Medicare FFS inpatient claims from VHCURES, calendar years 2019-2022, MPR analysis, 4. NVRH analysis using codes published in "Methods & Measures Used in the Reporting for Blueprint's Community Profiles"

REDUCING POTENTIALLY AVOIDABLE IP STAYS AND ED VISITS WILL ALLEVIATE CAPACITY CHALLENGES, BUT ALSO REDUCE REVENUE

Illustrative revenue impact on 2022 financials from potentially avoidable ED visits and IP admissions
$MN

Potentially avoidable IP stay

1,095 IP stays	×	**20%** avoidable	×	**50%** avoided	×	**$15,673** average reimbursement per IP stay	=	**$1.7 MN** reduction in revenue

Potentially avoidable ED visits

MPR analysis using NYU ED algorithm:

$106 MN Net patient revenue	×	**10.7%** from ED	×	**32.7%** avoidable	×	**50%** avoided	=	**$1.8 MN** reduction in revenue

NVRH analysis using Blueprint codes:

$106 MN Net patient revenue	×	**10.7%** from ED	×	**9.6%** avoidable	×	**50%** avoided	=	**$0.5 MN** reduction in revenue

REDUCING POTENTIALLY AVOIDABLE IP STAYS COULD ALSO FREE UP INPATIENT BED CAPACITIES OR MAY REDUCE VARIABLE COSTS

2022 NVRH discharges and costs associated with avoidable IP stays[1]

Assuming 50% potentially preventable IP stays are avoided:

		Discharges	Bed days	Potential staffed beds 'free-able'[2]
Total IP	-	1,095	-	-
Preventable	20% *of total*	219	750	-
Avoided	50% *of preventable*	110	375	**1.3**

Likely an under-estimate due to incomplete capture of claims data in VHCURES

Note: Avoidable IP impact on bed capacity and cost will be considered later

1. Avoidable hospital use refers to preventable hospitalizations and unplanned 30-day readmissions after an inpatient discharge. Preventable hospitalizations were identified using the Prevention Quality Overall Composite Indicator #90 (PQI #90), developed by the Agency for Healthcare Research and Quality. 30-day unplanned readmissions were identified using the HEDIS Plan All-Cause Readmissions (PCR) measure developed by the National Committee for Quality Assurance. The data above aggregate acute inpatient claims into stays by combining overlapping claims or clams indicating transfers to a different hospital for continuing care. Preventable hospitalizations with readmissions were combined to arrive at deduplicated counts for total avoidable hospital use. Since some preventable hospitalizations are also unplanned readmissions and vice versa, duplications were removed to avoid double-counting. 2. Assuming 80% bed occupancy rate, 365 days in a year
Source: Commercial, Medicaid and Medicare FFS inpatient claims from VHCURES, calendar years 2019-2022, MPR analysis

4.3

POTENTIALLY AVOIDABLE BOARDERS

NVRH COULD SAVE $0.5-2.2MN IN OPERATING COST IF BOARDER STAYS CAN BE REDUCED

2022 NVRH discharges and costs associated with 'boarders'[1,2]

	Discharges	Bed days[3]	Potential staffed beds 'free-able'[4,5]
IP	193	229	**0.8**
ED	1,792	2,412	**2.1 - 8.3**

Note: IP boarder impact on capacity and cost will be considered later

✖ **~$900** average variable cost per ED boarder day

$0.5MN -$2.2MN potentially avoidable variable cost from ED boarder reduction[3]

2022 NVRH Outpatient ED Discharges and Days by Discharge Status for Patients with a Mental Health Related Diagnosis

Discharge Status	Discharges	Days	Average LoS
01, Home or Self Care	235	139	0.6
02, Other Acute Hospital	54	163	3.0
05, Other Health Care Facility	19	69	3.6
07, Against Medical Advice	8	12	1.5
30, Still Patient	6	9	1.5

2022 NVRH Outpatient ED Discharges and Days for Visits without a Same Day Discharge by Discharge Disposition

Discharge Status	Discharges	Days	Average LoS
01, Home or Self Care	1,046	1,371	1.3
02, Other Acute Hospital	90	173	1.9
03, Skilled Nursing Facility	11	21	1.9
05, Other Health Care Facility	16	78	4.9
06, Organized Home Health Service	50	94	1.9
07, Against Medical Advice	21	28	1.3
30, Still Patient	13	19	1.5

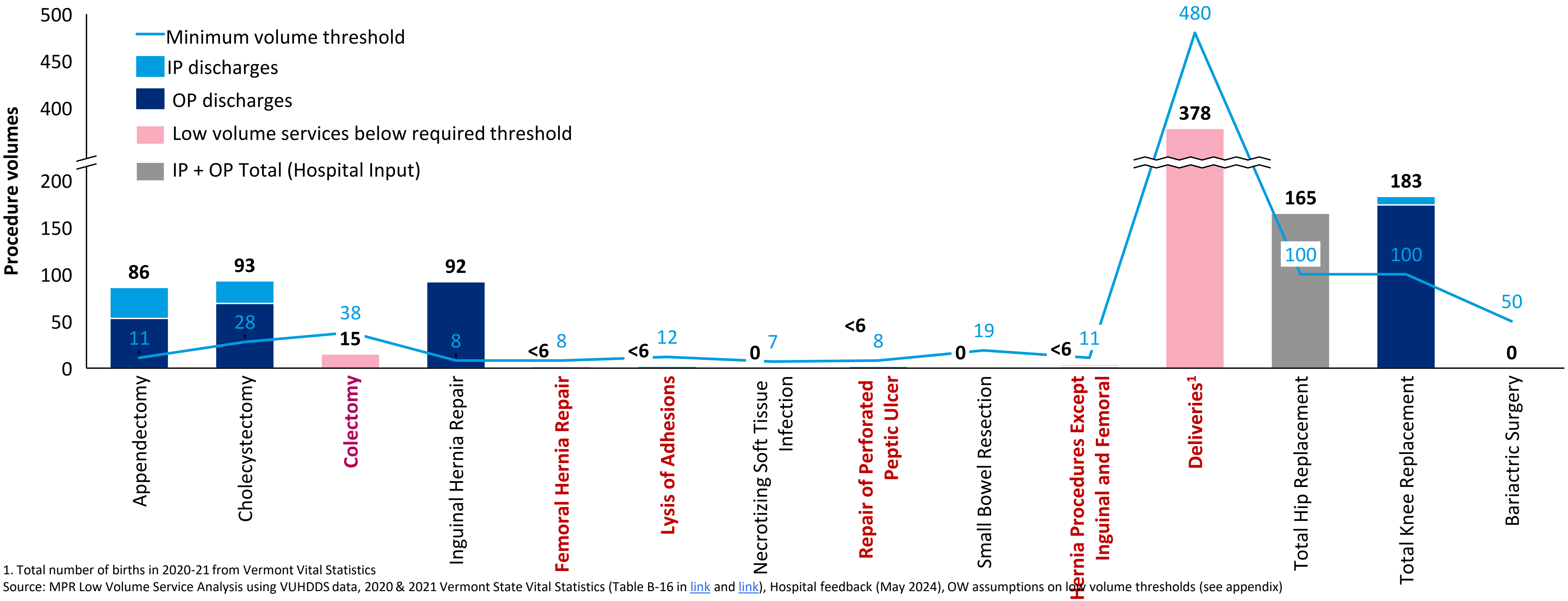
1.'Boarder' analysis included ED discharges and days for visits without a same day discharge, inpatient and outpatient ED discharges and days with at least 1 SDOH Z-code, inpatient and outpatient ED discharges and days for patients with a mental health related diagnosis, and outpatient ED discharges and days by discharge status for potential long-term care discharge statuses. 2. VAHHS discharge data uses a zero-day methodology for calculating LOS, meaning a patient who arrives and leaves the same day has a LOS of zero. 3. LOS for inpatient ED visits is an estimate derived off of total hospital LOS minus reported inpatient room and board units. 4. Assuming 80% bed occupancy rate, 365 days in a year, 5. For ED, lower bound is determined assuming average stays beyond 1 night at the ED are boarder stays that can be reduced, upper bound is determined assuming all over-night stays in the ED can be reduced. Source: VAHHS analysis using 2022 hospital discharge data (Discharge values for any category that were less than 5 have been excluded), Oliver Wyman analysis

4.4

LOW VOLUME PROCEDURES

SOME PROCEDURES PERFORMED AT NVRH TODAY MAY NOT HAVE ENOUGH VOLUMES TO ENSURE COMPETENCIES ARE RETAINED AND LIMIT PATIENT SAFETY RISK

Volume of procedures performed at NVRH and minimum volume standards to ensure care quality
2021-2022 total



1. Total number of births in 2020-21 from Vermont Vital Statistics
Source: MPR Low Volume Service Analysis using VUHDDS data, 2020 & 2021 Vermont State Vital Statistics (Table B-16 in link and link), Hospital feedback (May 2024), OW assumptions on low volume thresholds (see appendix)

TREATMENT OPTIONS FOR LOW VOLUME SURGERY

Response Times for IVR and Endoscopy at Center for Emergency General Surgery[1]

Condition	Time from Request
Sepsis source control (shock)	60 minutes
Sepsis source control (no shock)	3 hours
Biliary drainage (endoscopic retrograde cholangiopancreatography [ERCP] or percutaneous transhepatic cholangiography [PTC]) for biliary sepsis/ cholangitis (not in shock)	12 hours
Bleeding (ongoing or shock)	60 minutes
Intermittent bleeding, hemodynamically stable	12 hours, if needed
Complete large bowel obstruction (closed-loop) with signs of peritonitis	60 minutes
Complete large bowel obstruction (closed-loop) without signs of peritonitis	3 hours

Potential options for low volume procedures

Elective cases	Stop doing and transfer volumes to other hospitals
	Grow volume to meet required threshold to ensure care quality
Emergent cases	Transfer to a regional center with capability
	For extreme cases, do locally with appropriate consent

Source: 1. ACS 2022

4.5

‘RE-CAPTURE’ VOLUMES

WE IDENTIFIED INPATIENT AND OUTPATIENT VOLUMES WITH 'RE-CAPTURE' POTENTIAL BY SCREENING FOR RESIDENT ORIGIN, PROCEDURE COMPLEXITY AND SPECIALTY AVAILABILITY

Methodology Steps[1]

Identified hospital discharges outside of a patient's home HSA as being eligible to return to the home HSA, except those that require tertiary care / highly specialized equipment or expertise (see appendix for details)

Removed services intended by PCP providers and services that would go to UVM[1]

Removed discharges <11 due to uncertain volume

Only services your hospital has physician specialist/s for

Assumptions & Implications:

1. Subsequent calculations do not take into account the number of physicians in each specialty, i.e. we assumed all qualifying discharges with 're-capture' potential can be serviced in their local hospital, without specialist staffing capacity constraints
2. Total estimated IP and OP volumes with re-capture potential are likely an under-estimate due to omission of discharges with n<11

Current/intended physician resourcing[2,3]

Specialty List
Cardiology
ENT
General Surgery
GYN
Neurology
Internal Medicine[2]
OB
Orthopedics
Podiatry
Pulmonary
Urology

1. For Inpatient data, additional complex Neonatology and Cardiovascular Surgery procedures were out of scope due to patients going to UVM/Dartmouth; For outpatient data, key Transplant (kidney, liver, lung, bone marrow) and stem cell therapies were considered to be done by UVM 2. All hospitals were assumed to have internal medicine capabilities 3. Specialty list provided by hospital feedback (May 2024)

Source: MPR Final Inpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data, MPR Final Outpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data, Hospital feedback (May 2024), OW analysis

WE USED THE FOLLOWING HOSPITAL CAPACITY NUMBERS IN SUBSEQUENT CAPACITY CALCULATIONS

Current hospital capacity by category

Capacity	Number
Licensed Beds	25 (CAH status, total licensed is 69)
Staffed Beds	21
Birthing Unit	4
Mental Health beds	0
ICU	4
IP Rehab	0
Swing/Progressive beds	
ED bays	9[1]
ED Mental Health Beds	4[1]
NICU	
Operating room	3[1]
Procedure room	1[1]

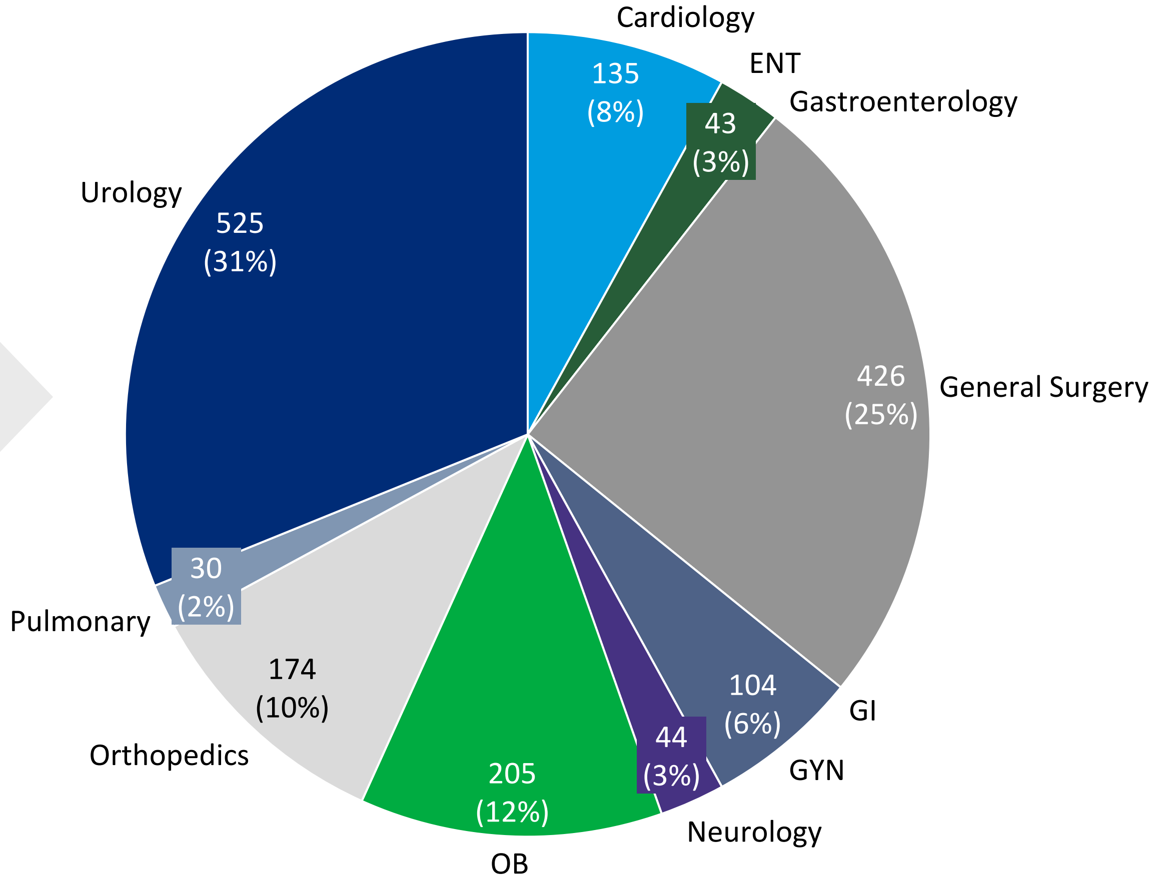
1. OW interviews (2023, 2024)
Source: MPR Final Outpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data , Hospital feedback (May 2024), 2022 CMS payment rates, OW Analysis

GIVEN NVRH PHYSICIAN SPECIALTIES, ~1680 OUTPATIENT DISCHARGES IN 2022 HAD THE POTENTIAL TO BE RE-CAPTURED

Volume of outpatient services with 're-capture' potential, by specialty
2022

Total Number of Discharges

1686



Remarks

- Urology had the most overall discharges across all diagnoses with 1354 discharges
- Urology also had the maximum discharge for a specific diagnosis, with 226 discharges under malignant prostate neoplasms
- Highest General surgery services were related to unspecified abdominal pain

Discharge breakdown by procedure type	
Ambulatory	0 (0%)
OR	43 (3%)
PR	1643 (97%)

Source: MPR Final Outpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data, OW analysis. See appendix for more detail on methodology

WE ANALYZED THE CAPACITY AND FINANCIAL IMPLICATIONS OF OUTPATIENT SERVICES WITH 'RE-CAPTURE' POTENTIAL BY THEIR SERVICE SETTING

Categorization of outpatient services with 're-capture' potential by service setting

	Operating room (OR)	Procedure room (PR)	Ambulatory clinic
Methodology for classification	• Procedure types that were longer in duration (avg. 2hrs) and required full OR set-up and equipment • Identified through fuzzy-match with relevant specialities[1]	• Procedure types that were shorter in duration (avg. 30min) and required less complex equipment • Identified through fuzzy-match with procedures related to ENT or Ophthalmology	• All services that would not need specialized operating or procedure rooms • Identified through fuzzy-match with diagnoses associated with Endocrinology, Infectious Diseases, Nephrology, Hematology, Oncology, Podiatry, SUD
Calculations on capacity and financial impact	• Estimated using current OR capacity and 2022 median hospital revenue per OR-based OP procedure	• Estimated using current PR capacity and 2022 median hospital revenue per PR-based OP procedure	• Not estimated

Note: ongoing CMS push to align hospital payments to office-based payments would likely reduce expected revenue over time for ambulatory clinic services, potentially off-setting financial impact of volume uplifts

1 Specialists included Anesthesia, Cardiology, Dermatology, Gastroenterology, General Surgery, GYN, Neurology, Neurosurgery, OB, Thoracic Surgery, Urology, Orthopedics, Rheumatology, Pulmonary, Psychiatry, Plastic Surgery, & Pediatrics
Source: MPR Final Outpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data, 2022 CMS payment rates, OW Analysis

ACTUAL NUMBER OF OUTPATIENT VOLUMES RE-CAPTURED WOULD DEPEND ON SURGICAL CAPACITY, CURRENT THROUGHPUT, STAFF CAPACITY AND PATIENT PREFERENCE

Current surgical capacity
2023

3
Operating Rooms

40 hrs per week × **48** weeks per year ÷ **2** hrs per procedure = **5,760** procedures per year

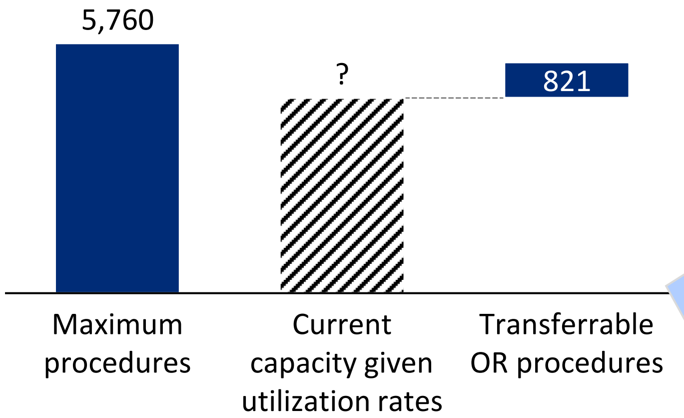
1
Procedure Room

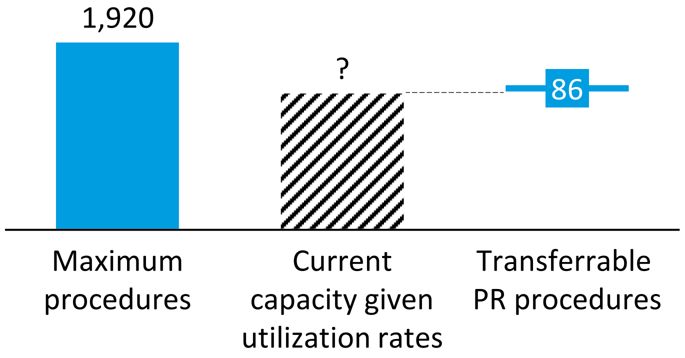
40 hrs per week × **48** weeks per year ÷ **0.5** hrs per procedure = **1,920** procedures per year

Total Number

7,680
procedures per year

Maximum capacity for number of procedures[1-3]
2023





Discuss: to what extent can the potential change in OP volume
- ***Be handled within capacity***
- ***Affect fixed / variable cost?*** *(current assumption is 98% of additional revenue are cost)*

1. Current OR and PR utilization rates are unknown and influence the capacity for number of procedures able to be transferred back 2. Transferrable outpatient services were separated into ambulatory services, OR services, and PR services. 3. Unknown cost estimates and utilization rates makes revenue calculations not currently feasible
Source: MPR Final Outpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data, Hospital feedback (May 2024), Oliver Wyman Analysis

NVRH MAY BE ABLE TO RE-CAPTURE SOME INPATIENT CARE CURRENTLY SOUGHT OUTSIDE OF ST. JOHNSBURY

Number of 2022 inpatient discharges of patients from St. Johnsbury that could return to NVRH by DRG[1,2]

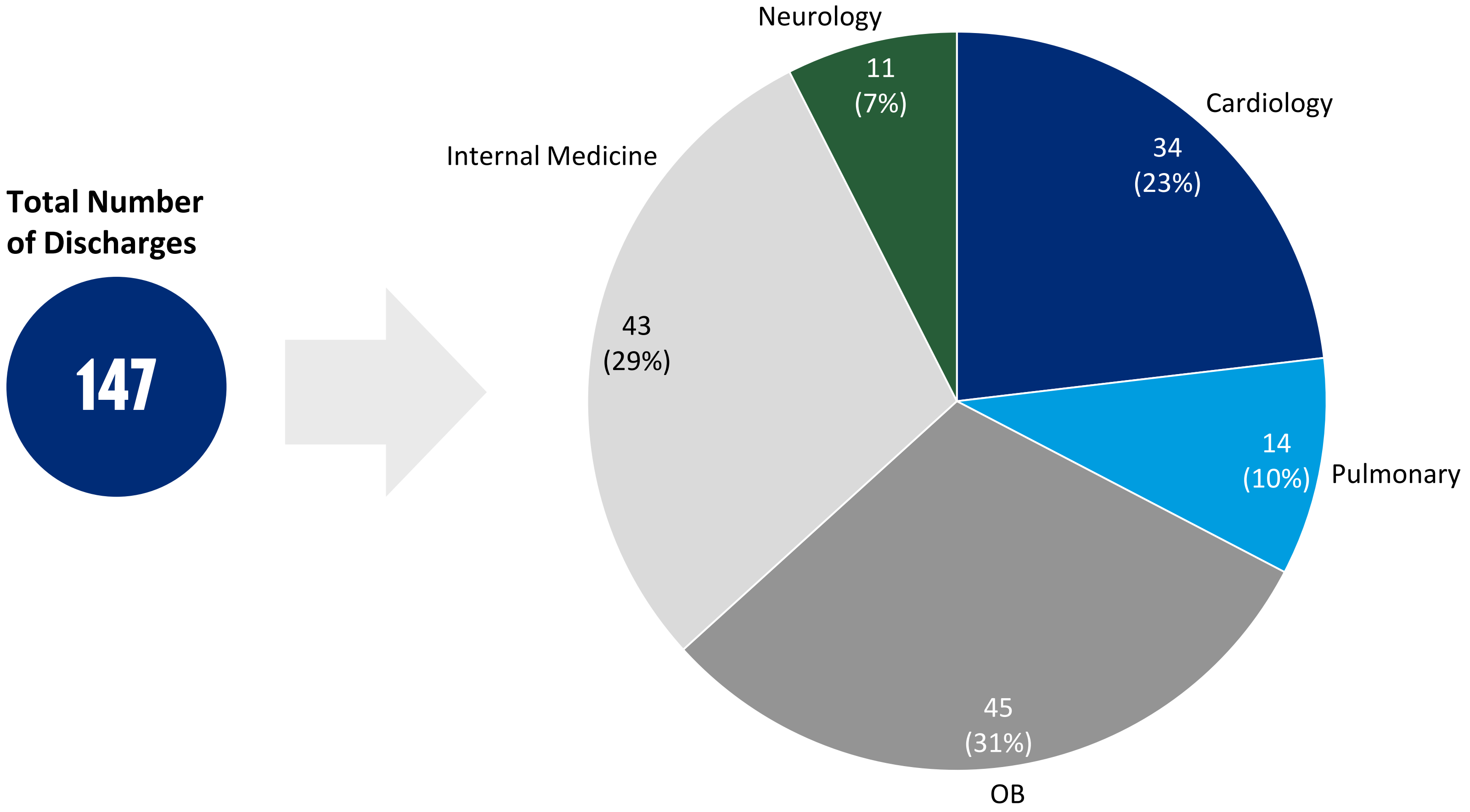
Note: Only qualifying IP discharges with n≥11 are shown below and included in subsequent calculation

DRG	DRG Description	Number of Discharges	Considered re-capturable based on hospital-indicated specialty availability?
885	PSYCHOSES	95	No
871	SEPTICEMIA OR SEVERE SEPSIS WITHOUT MV >96 HOURS WITH MCC	24	Yes
807	VAGINAL DELIVERY WITHOUT STERILIZATION OR D&C WITHOUT CC/MCC	22	Yes
897	ALCOHOL, DRUG ABUSE OR DEPENDENCE WITHOUT REHABILITATION THERAPY WITHOUT MCC	19	Yes
881	DEPRESSIVE NEUROSES	16	No
177	RESPIRATORY INFECTIONS AND INFLAMMATIONS WITH MCC	14	Yes
309	CARDIAC ARRHYTHMIA AND CONDUCTION DISORDERS WITH CC	12	Yes
795	NORMAL NEWBORN	12	Yes
280	ACUTE MYOCARDIAL INFARCTION, DISCHARGED ALIVE WITH MCC	11	Yes
788	CESAREAN SECTION WITHOUT STERILIZATION WITHOUT CC/MCC	11	Yes
291	HEART FAILURE AND SHOCK WITH MCC	11	Yes
64	INTRACRANIAL HEMORRHAGE OR CEREBRAL INFARCTION WITH MCC	11	Yes

Note: 1. Inpatient discharges were identified in VHCURES using Inpatient Discharge ID. Discharges at hospitals outside of a patient's home HSA were identified as being eligible to return to the hospital in the patient's home HSA, if – (1) the admission was for a DRG that did not require tertiary care, or (2) the patient's home HSA was Burlington in which case UVMMC was assumed to be able to care for all DRGs, including tertiary care DRGs. DRGs that require a higher level of specialty care and highly specialized equipment or expertise were identified by Mathematica as tertiary care DRGs on prior work for GMCB, based on DRG weights. All HSAs except Brattleboro had only a single hospital in the HSA. 2. Number of discharges less than 11 are masked and not shown in the table above.
Source: MPR Final Inpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data, OW analysis

SOME INPATIENT SERVICES MAY ALSO BE RECAPTURED AT NVRH, AND ARE CONCENTRATED IN INTERNAL MEDICINE, OB, CARDIOLOGY, AND PULMONARY SPECIALTIES

Volume of inpatient services with 're-capture' potential, by specialty



Remarks

- After internal medicine sepsis (24), OB Vaginal delivery had the most inpatient discharges (22)
- Cardiology related services related most commonly to arrythmia and conduction disorders (12), myocardial infarction (11) and heart failure (11)
- Respiratory infections and inflammations with MCC was the only Pulmonary-related DRG code with ≥ 11 discharges captured

1. All hospitals were assumed to have internal medicine capabilities
Source: MPR Final Inpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022, Hospital Data provided, Oliver Wyman Analysis. See appendix for more detail on methodology

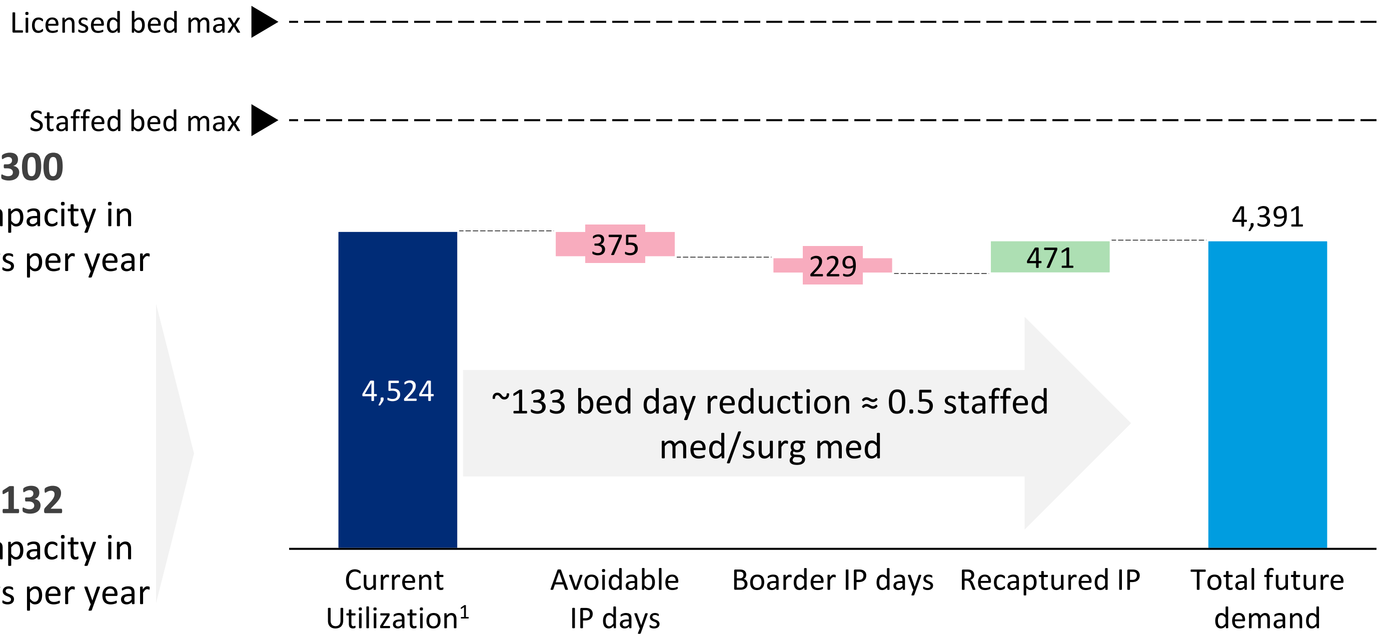
FOR IP SERVICES, NVRH SHOULD HAVE THE BED CAPACITY TO DEAL WITH FUTURE DEMAND AFTER AVOIDABLE, BOARDER & POTENTIALLY RE-CAPTURED SERVICES ARE ACCOUNTED FOR

Current maximum licensed and staffed bed capacity (supply)
2023, bed days

25 Licensed beds — **365** days ✖ **80%** target occupancy = **7,300** Max capacity in bed days per year

21 Staffed med-surg beds — **365** days ✖ **80%** target occupancy = **6,132** Max capacity in bed days per year

Current and potential future IP med/surg bed demand
2022, bed days



Discuss: to what extent would the potential change in IP volume affect variable cost?

1 Current utilization based on average of 2018-2021 IP discharges and LoS from 2021 Vermont Hospital Report
Source: Hospital feedback (May 2024), [Current Utilization] 2021 Vermont Hospital Report, [Avoidable IP days] Commercial, Medicaid and Medicare FFS inpatient claims from VHCURES, calendar years 2019-2022, MPR analysis, [Boarder] VAHHS analysis using 2022 hospital discharge data (OW Survey CY 2022), [Transferred IP] MPR Final Inpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data, OW Analysis

RECAPTURED IP AND OP SERVICES HAVE A MODEST OPPORTUNITY TO INCREASE REVENUE

Illustrative revenue impact on 2022 financials from potentially recapturable IP and OP discharges
$ USD, 2022

OP discharges with 're-capture' potential

~576 Operating room based discharges × $4,506 per procedure[1]

43 Procedure room based discharges × $1,124 per procedure[1]

= $2.6 MN increase in revenue

+

IP discharges with 're-capture' potential

135 discharges × $15,673 average reimbursement per IP stay[2]

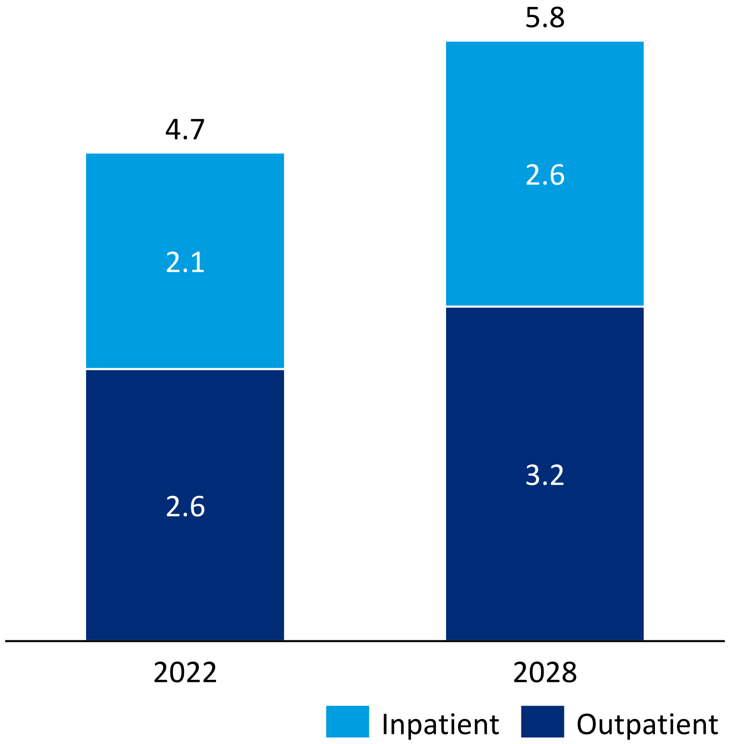
= $2.1 MN increase in revenue

=

Total

$4.7 MN increase in revenue

Estimated net revenue impact on recapturable IP and OP discharges[3]
$MN



1. OP revenue per procedure was calculated as estimated median for select operating room based procedures and median for select procedure room based procedures from 2022 CMS payment rates 2. Inpatient revenue was calculated as the average of the calculated net average revenue per inpatient discharges 2018-2021 3. Projected revenue is based on 2022 data and assumed increases of cost of 3.5% annually

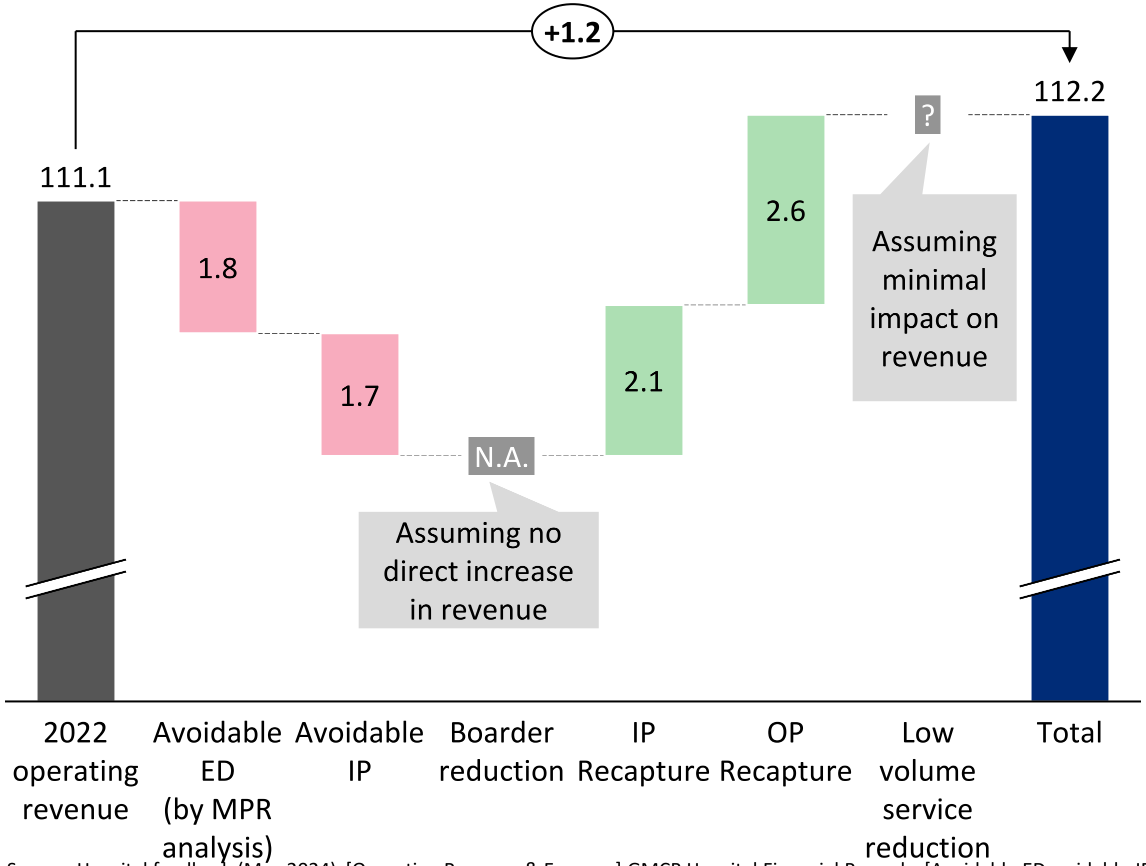
Source: MPR Final Inpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data, MPR Final Outpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data, Avoidable hospital use analysis by MPR using Commercial, Medicaid and Medicare FFS inpatient claims from VHCURES, calendar years 2021-2022, 2022 CMS payment rates, OW Analysis

4.6

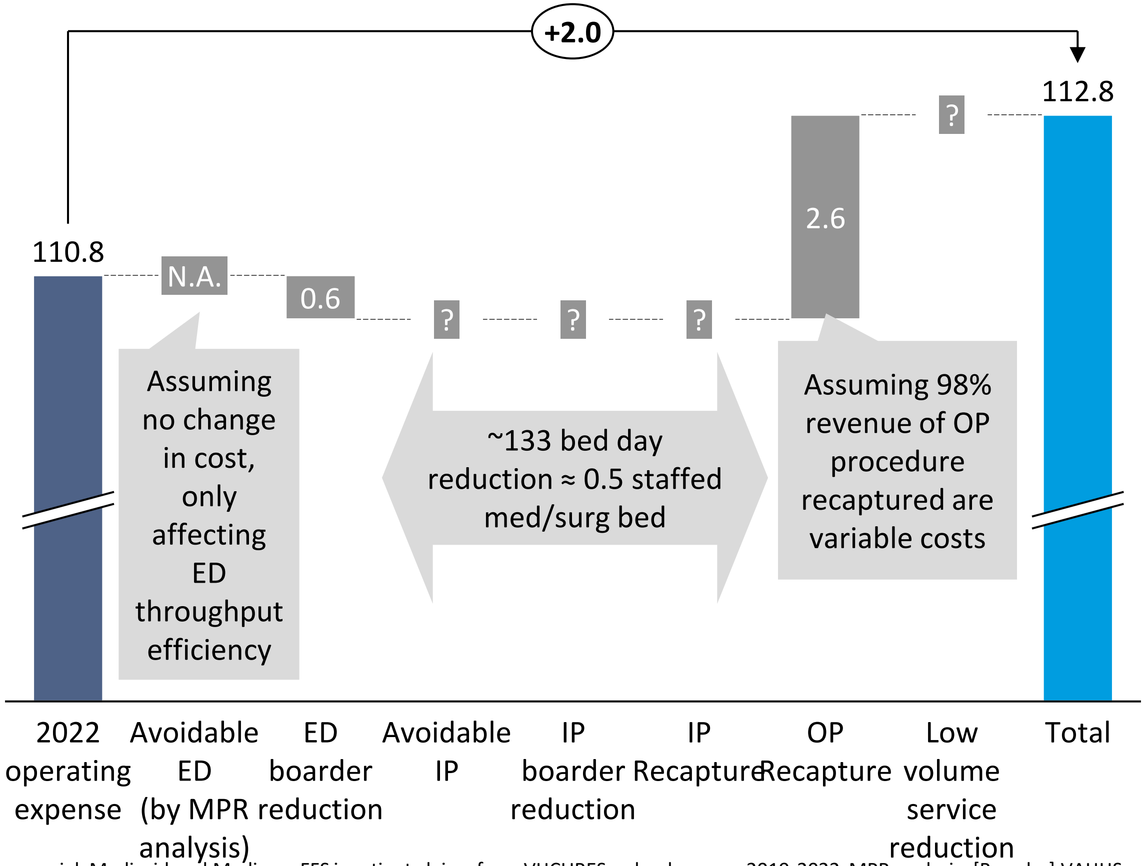
TOTAL IMPACT

OVERALL FINANCIAL IMPACT IS MOST INFLUENCED BY OP RECAPTURED SERVICES FOR BOTH OPERATING REVENUE AND EXPENSES

Illustrative financial impact on 2022 operating revenue
$USD MN



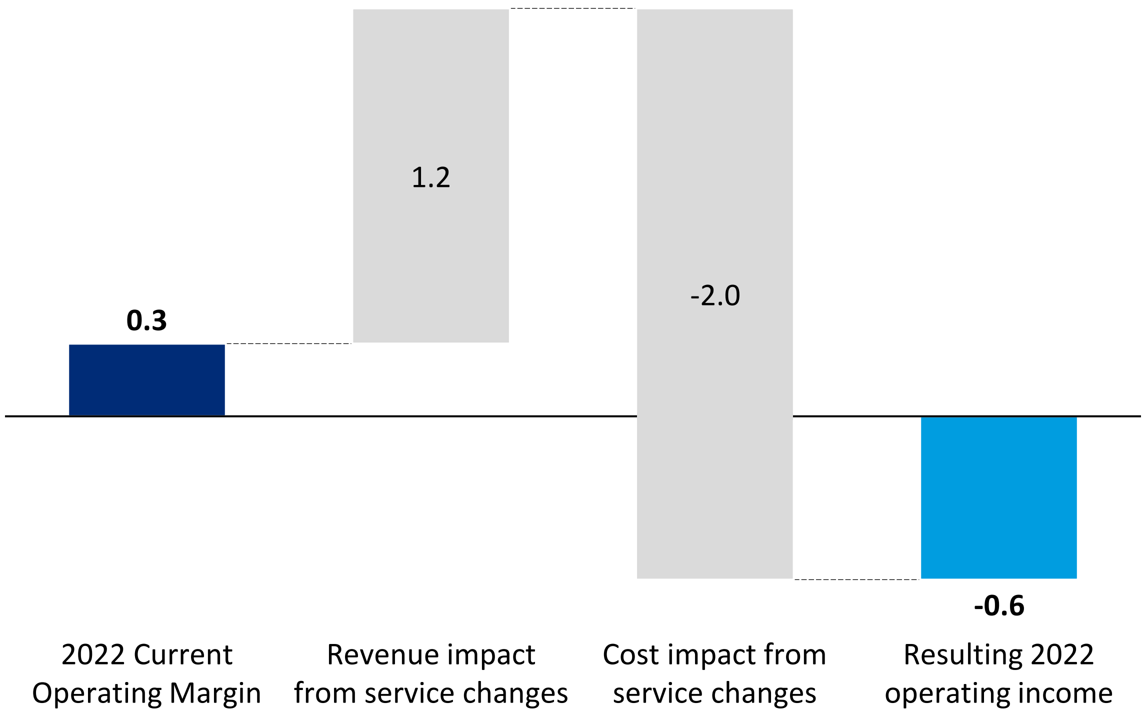
Illustrative financial impact on 2022 operating expense
$USD MN



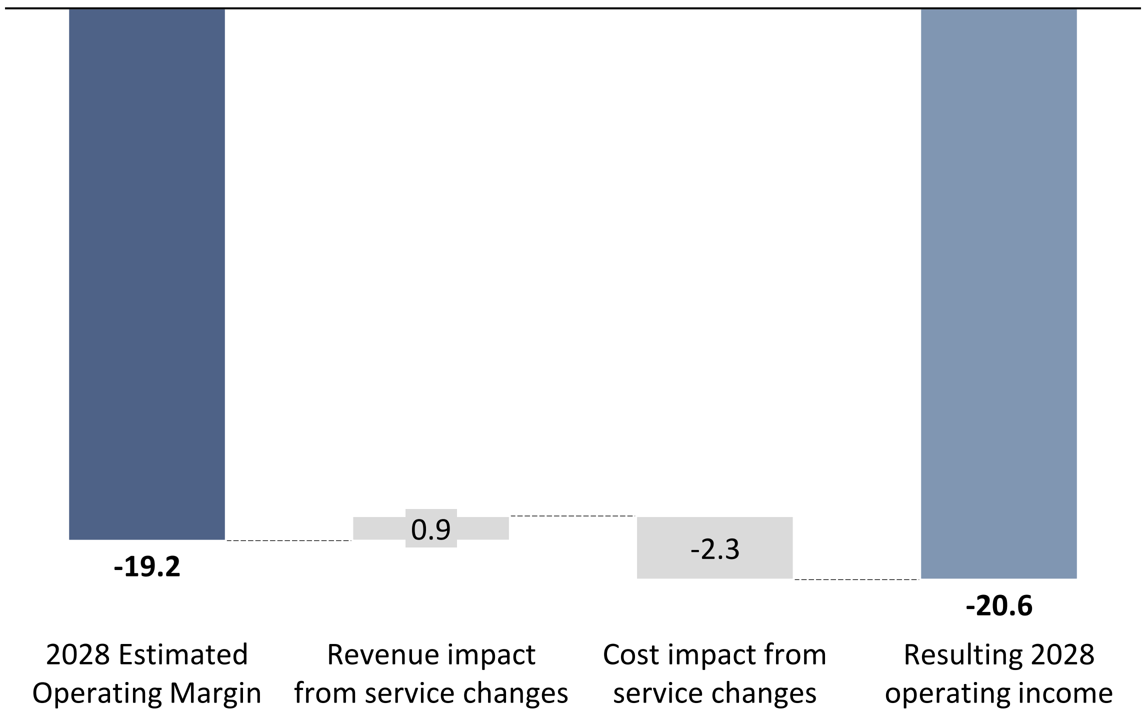
Source: Hospital feedback (May 2024), [Operating Revenue & Expense] GMCB Hospital Financial Records, [Avoidable ED, voidable IP days] Commercial, Medicaid and Medicare FFS inpatient claims from VHCURES, calendar years 2019-2022, MPR analysis, [Boarder] VAHHS analysis using 2022 hospital discharge data (OW Survey CY 2022), [Transferred IP] MPR Final Inpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data, [Transferred OP] MPR Final Outpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data, [Low Volume] MPR Low Volume Service Analysis using VUHDDS data, 2020 & 2021 Vermont State Vital Statistics (Table B-16 in link and link), OW assumptions on low volume thresholds (see appendix) OW Analysis

DESPITE POTENTIAL ADJUSTMENTS, OPERATING INCOMES FALL WELL BELOW ZERO, DEMONSTRATING MINIMAL OVERALL IMPACT AND NEED FOR LARGER STRUCTURAL CHANGE

Illustrative financial impact on 2022 margin
$USD, MN



Illustrative financial impact on 2028 margin[1]
$USD, MN



1. Based on OW Assumptions of future growth in revenue and expenses
Source: GMCB hospital financial records, Oliver Wyman analysis

5. HOSPITAL OPTIONS

NORTHEASTERN VERMONT REGIONAL HOSPITAL OPTIONS

Initially proposed by OW + hospital feedback (June 2024)

RESULTS OF ANALYSIS

Hospital only

- Stop or grow colectomy
- Stop doing femoral hernia repair, lysis of adhesions, perforated ulcers and other hernia
- Grow OB or shift to other organizations

Cross-hospital *(unlikely to improve financials)*

- Execute on shared services with other hospitals
- Strongly consider formation of a regional specialty medical group with other hospitals
- Expand telehealth support for emergency room/ UrgiCare and specialists

POSSIBILITIES/ OPTIONS

Possibilities

- **Develop joint venture** free-standing birthing unit with North Country hospital and/or Gifford Medical Center
- **Add PACE program** (pending population needs)
- **Develop rural outreach** programs for primary care and preventative services
- **Provide pharmacy services to community** through primary care sites with 5+ providers - ***ongoing***
- **Expand** infusion center with telehealth support
- **Expand home dialysis program**

Already ongoing efforts

- **Increase capacity** for cardiology (non-invasive) and surgical sub-specialties **-** ***ongoing***
- **Add** additional telehealth specialty consultations with referral centers **-** ***ongoing***
- **SASH program** targeted at high needs groups/individuals **–** ***ongoing***

6. OTHER RECOMMENDATIONS

NORTHEASTERN VERMONT REGIONAL HOSPITAL- OW FINAL RECOMMENDATIONS

Solutions for Interim Period (2025- 2027)

Northeastern Vermont Regional Hospital	• Stop performing femoral hernia repair, lysis of adhesions, perforated ulcers and other hernia procedures • Stop or grow colectomy • Expand telehealth support for emergency room/ UrgiCare and specialists
Cross-Hospital	• Combine all back office and support functions into New England Collaborative Network with other CAH and unaligned PPS hospitals in Vermont as in Cross Hospital • Form a consortium with North Country Hospital and Gifford to support a care in the home model and “Hospital At Home” for eligible patients; expand telehealth services and collaborate with other organizations to develop “hospital at home” model • Use consortium to provide mobile rural clinics / services in combined HSA • Strongly consider formation of a regional specialty medical group with other hospitals
Community-Based	• Expand mobile health clinic to Newport HSA • Collaborate with local EMS and neighboring hospitals to develop plans to coordinate regional EMS services • Explore programs aimed at serving high risk patients e.g., PACE / SASH

Solutions for Long-Term Period (2028+)

Northeastern Vermont Regional Hospital	• Start inpatient dialysis unit if / when Newport closes inpatient operations • Grow OB or shift to other organizations (e.g., OB services from Newport or northern part of White River Junction HAS) • Consider growing surgical services

Other Considerations

HSA Reconfigurations	• Combine St. Johnsbury and Newport HSAs, extend HSA south to incorporate part of northern part of White River Junction HSA

OTHER HOSPITAL-LEVEL COST REDUCTION INITIATIVES TO CONSIDER

Recommended action	Sub-category	Descriptions / detailed options	Rationale / impact
Seek operational synergy	**Improve EMR functionality**	Speed up adoption of VITL and embed into hospital workflow	✓ Improve efficiency of patient care ✓ Reduce duplication of testing
Seek cost synergy	**Pursue group purchase**	Pursue group purchasing (supplies, drug purchasing, insurance and group employee benefits) - *Ongoing*	✓ Reduce hospital operating expense
		Consider group purchase of equipment, services on equipment and common IT system	
	Centralize services	Centralize interpretative and linguistic services across all agencies with one phone number and single website	*Cost-benefit analyses should be conducted to ascertain whether centralization creates savings in the long-run when all direct and indirect costs are considered*
		Centralize laundry services and/or kitchen to prepare flash frozen meals for delivery to hospitals, SNFs, adult day care	
		Centralize central sterile supply for operating rooms	
		[Northwestern, Southwestern and Rutland only] Consider sub-contracting dietary and house-keeping services	
		Build a regional ambulatory surgery centers with 4-6 operating/procedure rooms and a recovery area to replace aged small inpatient ORs	✓ Maintain patient access to essential yet low-volume services whilst allowing for potential specialization ✓ Reduce staff cost
	Share staffing	Develop mobile health services with resources and cooperation between several HSAs	
		Allow smaller hospitals to form a corporation to jointly employ a "regional physician group" (esp. medical specialists) which could rotate MDs among the locations and provide internal telehealth support to the EDs	
		Share executive staff, operational staff (e.g. HR, quality, Infection Control, etc) and IT security staff between small hospitals	✓ Reduce staff cost

QUALIFICATIONS, ASSUMPTIONS, AND LIMITING CONDITIONS

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APPENDIX

METHODOLOGY FOR CALCULATING INPATIENT AND OUTPATIENT DISCHARGES WITH RE-CAPTURE POTENTIAL

Outpatient Discharge Methodology

- Outpatient discharges were identified in VHCURES using unique member ID, date of service, and the billing provider. The data table is based on the patient flow analysis for 2022 using VHCURES data that provides the number of discharges by hospital, patient's home HSA, and diagnosis.
- Outpatient hospital discharges outside of a patient's home HSA as being eligible to return to the home HSA, if:
  - The discharge was for a diagnosis that was identified by Oliver Wyman as being eligible to return (see below for more detail) OR
  - The patient's home HSA was Burlington in which case UVMMC was assumed to be able to care for all diagnoses (except dental, mental health, substance use disorder, and social service-related diagnoses that were considered avoidable).
- All HSAs except Brattleboro had only a single hospital in the HSA. For patients with Brattleboro as the home HSA, the number of returning discharges for a diagnosis was split between Brattleboro and Grace Cottage hospitals in a 80%/20% ratio, as long as discharges for a diagnosis were observed at Grace Cottage in 2022. Otherwise, all returning discharges for a diagnosis were assigned to Brattleboro hospital. Number of discharges less than 11 are masked.
- For a procedure to be eligible to return as determined by Oliver Wyman:
  - The hospital needs to have the assigned specialty for the procedure (as indicated on the right), which was provided by the hospital
  - Discharges numbers for a procedure had to be >11, and was otherwise considered too low
  - Services that Primary Care Physicians treat were not considered to be recapturable
  - Key Transplant (kidney, liver, lung, bone marrow) and stem cell therapies were considered to be done by UVM and not recapturable

Inpatient Discharge Methodology

- Inpatient discharges were identified in VHCURES using Inpatient Discharge ID.
- Discharge at hospitals outside of a patient's home HSA as being eligible to return to the hospital in the patient's home HSA, if
  - The admission was for a DRG that did not require tertiary care OR
  - The patient's home HSA was Burlington in which case UVMMC was assumed to be able to care for all DRGs, including tertiary care DRGs.
- DRGs that require a higher level of specialty care and highly specialized equipment or expertise were identified by Mathematica as tertiary care DRGs on prior work for GMCB, based on DRG weights. All HSAs except Brattleboro had only a single hospital in the HSA. For patients with Brattleboro as the home HSA, the number of returning discharges for a DRG was split between Brattleboro and Grace Cottage hospitals in a 80%/20% ratio, as long as discharges for a DRG were observed at Grace Cottage in 2022. Otherwise, all returning discharges for a DRG were assigned to Brattleboro hospital.
- Additional screening including:
  - Number of discharges less than 11
  - Services provided by Primary care specialists
  - Complex Neonatology and Cardiovascular Surgery procedures, which were considered out of scope due to patients going to UVM/Dartmouth

Source: MPR Final Inpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data, MPR Final Outpatient Return to HSA Discharges Analysis based on patient flow analysis for 2022 using VHCURES data, Hospital feedback (May 2024), OW analysis

REFERENCES FOR LOW VOLUME PROCEDURES

Procedure	
Appendectomy	(1)
Cholecystectomy	(1)
Colectomy	(1)
Inguinal Hernia Repair	(1) (4)
Femoral Hernia Repair	(1) (4)
Lysis of Adhesions	(1)
Necrotizing Soft tissue Infection	(1)
Repair of Perforated Peptic Ulcer	(1)
Small bowel Resection	(1)
Umbilical Hernia Repair	(1)
Ventral Hernia Repair	(1)
Deliveries	(2) (3)
Total joint replacement – hip	(1) (4)
Total joint replacement – knee	(1) (4)
Bariatric surgery	(5)

Reference

1) Leapfrog Hospital Survey-Inpatient Surgery, 2020

2) Obstetric Volume and Severe Maternal Morbidity Among Low-risk and Higher Risk Patients Giving Birth at Rural and Urban US Hospitals, Kozhimannil KB, JAMA Health Forum. 2023;4(6):e232110. doi:10.1001/jamahealthforum.2023.2110

3) Hospital Level Factors Associated with Anesthesia Related Adverse Events in Cesarean Deliveries, New York State 2009-2011. Guglielminotti J, Anesth. Analges. 2016; 122:1947-56

4) Selective Referral to High Volume Hospitals- Estimating Potentially Avoidable Deaths. Dudley RA, *et al*. JAMA 2000; 283:1159-1166

5) Requires certification as a Center of Excellence

RESPONSE TIMES FOR IVR AND ENDOSCOPY AT CENTER FOR EMERGENCY GENERAL SURGERY

Endoscopy/Interventional Radiology Availability

Condition	Time from Request
Sepsis source control (shock)	60 minutes
Sepsis source control (no shock)	3 hours
Biliary drainage (endoscopic retrograde cholangiopancreatography [ERCP] or percutaneous transhepatic cholangiography [PTC]) for biliary sepsis/ cholangitis (not in shock)	12 hours
Bleeding (ongoing or shock)	60 minutes
Intermittent bleeding, hemodynamically stable	12 hours, if needed
Complete large bowel obstruction (closed-loop) with signs of peritonitis	60 minutes
Complete large bowel obstruction (closed-loop) without signs of peritonitis	3 hours

Reference

- ACS 2022